Digital Transformation Handbook: Planning for Human Resources for Health and Health Worker Registries

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About this draft

Each chapter gives objectives, key messages, proposed sections with the main points to be covered, suggested figures and tables, tools and checklists, and the outputs that a country would be expected to produce. The content draws on published WHO guidance listed at the end of this document.

This is the working document of the handbook, seeded from the TWG manuscript outline of 2 October 2026. Working-group members comment on any block; comments are grouped into change sets, decided, and made on a branch with a preview of the result (see the repository README).

Questions for the TWG

- Does the five-part structure work for both health workforce and digital readers?
- Is anything missing that countries are asking for?
- Is anything included that does not belong in this handbook?
- Is the level of technical detail right for the main text, or should some of Part 2 move to annexes or online resources?

Points for discussion

Remark **One registry or one set of information?.** The WHO 2015 minimum data set describes the registry as “the single authoritative source”, and the draft DPI-H reference architecture appears to describe it as a centralized system. This draft treats the registry as one trusted set of information that may sit in one system or several. The TWG should agree on the wording.

Remark **Boundary with HRIS:.** The draft treats human resource management functions as out of scope. The TWG should confirm where the line is drawn, since many countries run the registry inside an HRIS.

Remark **Minimum data set:.** The draft proposes updating the 2015 minimum data set within this handbook. The TWG should confirm whether the update belongs here or in a separate product.

Remark **Unique health worker identification:.** Should this topic have a dedicated chapter?

Remark **Which health workers?.** The draft assumes regulated professions first, extending to other occupations including community health workers and the private sector. The TWG should confirm the scope.

Remark **Overlap between chapters:.** Governance appears in three chapters (Governance and Stewardship, Interoperability, Data Quality and Governance). Requirements appear in three (User Personas and Use Cases, Functional Requirements, Business Process Redesign). Use cases appear in two (User Personas and Use Cases, Illustrative Use Cases). Consider merging or cross-referencing, and renaming “Data Quality and Governance” to “Data Quality”.

Remark **Step-by-step or modular?.** Part 3 is drafted as a pathway that countries can enter at different points, taking the parts they need. Is this the right framing.

Remark **Case studies:.** No countries have been selected yet. The TWG should nominate cases.

Handbook at a glance

Table.

Part	Chapter	Sections
Part 0. Introduction	1. Why Digital Health Workforce Registries Matter	Global challenges; Value proposition; Registry as digital public infrastructure; Benefits and outcomes
	2. Scope and Boundaries	Included topics; Excluded topics; Target audiences
	3. Digital Transformation Approach	DIIG alignment; Transformation principles; Country pathway
Part 1. Foundations	4. Purpose and Role of a Registry	Definitions; Authoritative source; Health worker lifecycle; Strategic uses
	5. User Personas and Use Cases	Personas; Priority use cases; User stories; Requirements derivation
	6. Governance and Stewardship	Institutions; Legal foundations; Data stewardship; Sustainability
	7. Functional Requirements	Identity management; Records; Lifecycle

Part	Chapter	Sections
		management; Verification services; Analytics
	8. Digital Architecture Principles	Architecture patterns; Federation; Scalability; Technology neutrality
Part 2. Technical Design	9. Registry within Ecosystem	System landscape; Data flows; Architecture examples
	10. Interoperability	Standards; APIs; Data exchange patterns; Governance
	11. Data Standards and Minimum Dataset	Core dataset; Classifications; Metadata; NHWA mapping
	12. Privacy, Security and Trust	Privacy by design; Cybersecurity; Access control; Audit trails
	13. Data Quality Management	Quality dimensions; Validation; Monitoring
Part 3. Country Implementation	14. Preparing for Transformation	Leadership; Teams; Readiness; Stakeholder engagement
	15. Business Process Redesign	Current state mapping; Future workflows; Requirements
	16. Planning and Procurement	Options analysis; Procurement approaches; Risk management
	17. Implementation and Deployment	Migration; Testing; Rollout; Operations
	18. Change Management	Communications; Adoption; Capacity strengthening
	19. Monitoring and Continuous Improvement	KPIs; Maturity; Evaluation
Part 4. Examples and Resources	20. Illustrative Use Cases	Licensing; Verification; Emergency response; Mobility

Part	Chapter	Sections
	21. Country Case Studies	Centralized; Federated; Fragile settings; Lessons learned
	22. Tools and Templates	Checklists; Roadmaps; Assessment tools
Annexes	Minimum Data Set; Data Dictionary; Governance Template; User Personas; Functional Requirements Catalogue; Interoperability Models; Readiness Assessment; Maturity Model; Glossary; NHWA Mapping	

Handbook at a glance: parts, chapters and sections

Chapter 1. Why Digital Health Workforce Registries Matter

Part 0. Introduction. This part explains why the handbook exists, what it covers and does not cover, who it is for, and the approach it takes: helping countries move from where they are today towards digital, interoperable health workforce registries built on shared digital building blocks.

Chapter objectives. This chapter will help readers to:

- Explain the health workforce data problems that a digital registry helps to solve.
- Describe the value of a registry for health workers, regulators, employers, planners and the public.
- Position the registry as a shared, reusable part of a country's digital public infrastructure for health.

Key messages

- Countries cannot plan, deploy, pay or protect a workforce that they cannot count or verify.
- A registry is a trusted, shared set of core information about health workers. It does not have to be one single system.
- Most countries already hold the data in regulatory, payroll, human resources and training systems. The gap is that these systems do not connect.
- Investing once in a shared registry serves many uses: regulation, service delivery, planning, emergencies and reporting.

1.1 Global challenges

- WHO projects a global shortfall of about 11 million health workers by 2030, with uneven distribution within and between countries.
- Workforce data are spread across ministries, regulatory councils, employers, training institutions and the private sector, often on paper or in stand-alone systems.
- The results are duplicate and "ghost" records, unverified credentials, slow licensing, weak planning and difficulty reporting to National Health Workforce Accounts (NHWA).

1.2 Value proposition

- One trusted identity and record for each health worker, reused by every system that needs it ("capture once, use many times").
- Faster and cheaper administrative processes: registration, licence renewal, verification, deployment and payroll reconciliation.
- Better evidence for decisions on training, recruitment, distribution, retention and migration.

1.3 Registry as digital public infrastructure

- The draft WHO/ITU reference architecture for digital public infrastructure for health (DPI-H) lists the health workforce registry as a foundational registry, alongside client, facility and product registries.
- Shared building blocks (identity, registries, data exchange, terminology) are built once and reused by many applications.
- The registry is publicly governed, interoperable and based on open standards, so countries stay in control and avoid dependence on one supplier.

1.4 Benefits and outcomes

- For health workers: portable and verifiable credentials, simpler renewals, and recognition of training and continuing professional development.
- For the public and employers: confidence that practitioners are qualified and authorized to practise.
- For government: reliable workforce numbers for planning, financing, emergency response and international reporting (NHWA, the WHO Global Code of Practice).

Figures, tools and expected outputs

Suggested figures and tables

- Figure: the registry at the centre, with inputs (training institutions, regulators, employers) and outputs (verification, planning, reporting).
- Table: common workforce data problems and how a registry addresses each.

Implementation tools and checklists

- One-page advocacy brief template for senior decision-makers.
- Quick checklist: "Do we have a registry problem?"

Expected outputs for countries

- A shared problem statement agreed by health workforce and digital leads.
- A high-level vision for the national registry.

Chapter 2. Scope and Boundaries

Chapter objectives. This chapter will help readers to:

- Define what the handbook covers and what is left to other guidance.
- Clarify how a registry differs from, and relates to, human resources information systems (HRIS), regulatory, payroll and education systems.
- Identify the audiences and how each should use the handbook.

Key messages

- The handbook covers the registry as a shared foundation, not the full range of human resource management functions.
- It applies to all health workers a country chooses to include: usually regulated professions first, then other occupations, including community health workers.
- It is technology-neutral. It describes what a registry must do and exchange, not which product to buy.
- It builds on existing WHO guidance on NHWA, practitioner regulation and digital health, and does not replace it.

2.1 Included topics

- Purpose, governance, minimum data set, functional requirements and architecture options for a registry.
- Interoperability with regulators, employers, training institutions, the facility registry, identity systems and NHWA reporting.
- A phased country pathway from assessment to routine operation, for paper-based, partly digital and fully digital starting points.

2.2 Excluded topics

- Detailed human resource management functions (recruitment, performance appraisal, leave, payroll processing), which belong to HRIS and payroll systems.
- Design of regulatory policy, scopes of practice and accreditation standards, which are covered by WHO guidance on health practitioner regulation.
- Workforce planning and labour market analysis methods, and endorsement of specific software products.

2.3 Target audiences

- Decision-makers: ministry of health leadership, health workforce directorates, digital health units, regulators and professional councils.

- Implementers: enterprise architects, business analysts, software developers, data managers and implementing partners.
- Supporters: funders, professional associations, training institutions, private sector employers, and WHO regional and country offices.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: the registry and its neighbouring systems (HRIS, regulatory systems, payroll, education systems, facility registry).
- Table: in scope, out of scope, and covered by other guidance.
- Table: reader guide showing which chapters matter most for each audience.

Implementation tools and checklists

- Scope definition worksheet: which occupations, sectors, levels and functions are included in the first phase.

Expected outputs for countries

- A documented national scope statement for the registry (occupations, sectors, functions and phases).

Chapter 3. Digital Transformation Approach

Chapter objectives. This chapter will help readers to:

- Explain how the handbook applies the WHO Digital Implementation Investment Guide (DIIG) to registries.
- Describe the principles that guide a registry transformation.
- Outline a phased pathway that a country can follow, whatever its starting point.

Key messages

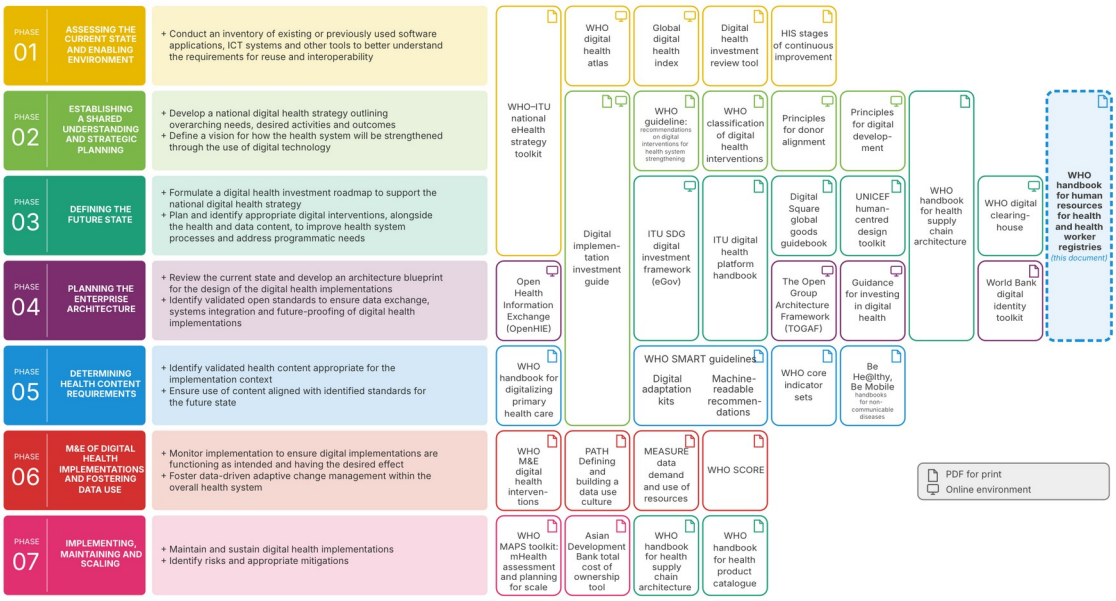
- Digital transformation changes processes, governance and skills. It is not only about installing software.
- Start from existing systems and improve them step by step. Full replacement is rarely necessary.
- Countries enter the pathway at different points and use the parts of the handbook that they need.
- Design for reuse from the start: standards, shared building blocks and open interfaces.

3.1 DIIG alignment

- The handbook follows the DIIG sequence: form the team, identify challenges, choose interventions, plan implementation, link to the enterprise architecture, budget and monitor.

- It sits alongside the other WHO digital transformation handbooks (primary health care, health supply chain architecture, health product catalogue) and the SMART Guidelines approach.
- It gives the health workforce detail that the DPI-H reference architecture does not go into.

Figure.



Source: adapted from WHO Digital transformation handbook for health product catalogue, 2025, Fig. 4, itself adapted from the 2020 WHO Digital implementation investment guide (DIIG), Fig. 1.1.1. Generated by the `diig-phases-svg` tool from the who-iris data for the product catalogue figure and this handbook's adaptation file; this handbook is added across phases 02–04 and marked "(this document)" in place of the product catalogue handbook.

Planning and implementing a digital health enterprise: phases, steps and resources, with this handbook in phases 02–04 (adapted from the product catalogue handbook, Fig. 4, and DIIG Fig. 1.1.1)

Table.

DIIG chapter	What the DIIG step produces	Where this handbook covers it for registries
2. Form the team and establish goals	Roles and responsibilities; a shared statement of needs and goals	Ch. 14 Preparing for transformation (14.1 Leadership, 14.2 Teams); Ch. 6 Governance and stewardship (6.1 Institutions); Ch. 2 Scope and boundaries
3. Identify health system	A map of the current	Ch. 1 Why registries

DIIG chapter	What the DIIG step produces	Where this handbook covers it for registries
challenges and needs	state; bottlenecks confirmed, prioritized and mapped to health system challenges	matter (1.1 Global challenges); Ch. 4 Purpose and role (4.2 Authoritative source); Ch. 14.3 Readiness; Ch. 15.1 Current state mapping
4. Determine appropriate digital health interventions	Selected interventions; enabling-environment assessment; functional requirements and user stories; a future-state workflow	Ch. 5 User personas and use cases; Ch. 7 Functional requirements; Ch. 15.2 Future workflows and 15.3 Requirements
5. Plan the implementation	Infrastructure; legislation, policy and compliance; leadership and governance; workforce and training; services and applications	Ch. 6.2 Legal foundations; Ch. 12 Privacy, security and trust; Ch. 17 Implementation and deployment; Ch. 18 Change management (18.3 Capacity strengthening)
6. Link to the enterprise architecture	Assessment of the architecture; common and enabling components; investment linked to both	Ch. 8 Digital architecture principles; Ch. 9 Registry within the ecosystem; Ch. 10 Interoperability; Ch. 11 Data standards and minimum data set
7. Develop a budget	Phases of implementation; cost drivers; a budget matrix	Ch. 16 Planning and procurement (16.1 Options analysis and total cost of ownership); Ch. 6.4 Sustainability
8. Monitor the implementation and use data effectively	A logic model; an M&E plan; a culture of data use; adaptive management	Ch. 13 Data quality management; Ch. 19 Monitoring and continuous improvement
9. Value proposition and next steps	The case made to whoever pays	Ch. 1.2 Value proposition and 1.4 Benefits and outcomes (with the advocacy brief

DIIG chapter	What the DIIG step produces	Where this handbook covers it for registries
		template)

DIIG chapter 1 (Introduction) corresponds to Part 0 of this handbook. Part 4 (Chapters 20–22: illustrative use cases, country case studies, tools and templates) supports every step. In the figure above (adapted from DIIG Fig. 1.1.1 by way of the product catalogue handbook), the DIIG spans phases 02–05, and this handbook sits across phases 02–04, adding the registry-specific detail within that span.

Source for the DIIG chapters and their outputs: WHO, ITU and UN Foundation, Digital implementation investment guide (DIIG), 2020, as summarized in `smart-base/methodologies/diig.md`*. The mapping is a proposal for the TWG.*

Handbook chapters mapped to the DIIG chapters (proposed, for TWG review)

3.2 Transformation principles

- Country-owned and publicly governed: meet countries where they are.
- Interoperable, standards-based, reusable, secure and privacy-protecting by design (in line with the draft DPI-H principles and the Principles for Digital Development).
- People-centred: designed with health workers, regulators and employers as users.

3.3 Country pathway

- Four broad stages: prepare and assess, design, implement, and operate and improve (detailed in Part 3).
- Typical starting points: paper registers; stand-alone digital systems in each council or ministry; partly connected systems.
- The guidance will also be published in a form that digital tools, including AI assistants, can use (for example machine-readable requirements and data dictionary).

Figures, tools and expected outputs

Suggested figures and tables

- Figure: country pathway showing entry points for different starting points.
- Table: handbook chapters mapped to DIIG steps.
- Figure: how this handbook relates to other WHO digital health and health workforce guidance.

Implementation tools and checklists

- “Where are we now?” self-placement tool (linked to the maturity model annex).

Expected outputs for countries

- An agreed transformation approach and guiding principles.
- The country's starting point and entry point on the pathway.

Chapter 4. Purpose and Role of a Registry

Part 1. Foundations. This part sets out the common understanding that health workforce and digital teams need before design starts: what a registry is and what it is for, who uses it, who governs it, what it must do, and the architecture principles it should follow.

Chapter objectives. This chapter will help readers to:

- Define a health workforce registry and related terms in plain language.
- Explain what "authoritative source" means and how it can be achieved with one system or several.
- Show how the registry supports each stage of the health worker life cycle and its main strategic uses.

Key messages

- The registry holds a minimum set of trusted information that uniquely identifies each health worker, their qualifications, their authorization to practise and where they work.
- "Authoritative" refers to the information and the rules for maintaining it, not to a single database.
- Each data item should have one accountable owner. The registry brings these items together.
- The registry is not an HRIS, but HRIS, regulatory and payroll systems should all supply data to it and draw on it.

4.1 Definitions

- Working definitions of health workforce registry, professional register, HRIS, health worker, registration, licence and credential (see Glossary annex).
- Alignment with the WHO 2015 minimum data set definition ("the single authoritative source of health workforce information") and with the draft DPI-H reference architecture.
- The difference between a "registry" (the shared national capability) and a "register" (a regulator's legal list of those authorized to practise).

4.2 Authoritative source

- Which institution is the source of truth for each data item, such as identity, qualification, licence status and employment.
- How records from several sources are matched to one unique health worker identifier.
- Options: a single national system, or several source systems connected to give one consistent view.

4.3 Health worker lifecycle

- Stages: pre-service education, graduation, registration and licensing, employment and deployment, continuing professional development, renewal, mobility, exit and re-entry.
- The events at each stage that create or change registry data, and who records them.

- Link to the NHWA health labour market framework (education, labour force, and serving population health needs).

4.4 Strategic uses

- Regulation and public protection: confirming that practitioners are qualified and in good standing.
- Planning and management: stock, distribution, skill mix, attrition and migration.
- Service delivery and digital health: identifying and authorizing health workers in electronic records, prescribing, telehealth and payment systems.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: health worker life cycle with the data events at each stage.
- Figure: “one set of information, one or many systems”.
- Table: registry compared with HRIS, professional register and payroll system.

Implementation tools and checklists

- Data ownership matrix template (data item by source institution).
- Terminology alignment checklist.

Expected outputs for countries

- An agreed national definition and purpose statement for the registry.
- A map of the authoritative source for each data item.

Chapter 5. User Personas and Use Cases

Chapter objectives. This chapter will help readers to:

- Identify the people and systems that supply or use registry information.
- Prioritize a small set of use cases to drive the first phase.
- Show how personas, use cases and user stories are turned into requirements.

Key messages

- Requirements should come from real users and tasks, not from software features.
- Systems are users too: many registry “users” are other applications requesting information through interfaces.
- Start with a few high-value use cases and add more over time.
- The generic personas and use cases are a starting point. Countries need to adapt them.

5.1 Personas

- People: health worker, registration officer in a regulatory body, employer or facility human resources manager, training institution registrar, workforce planner, member of the public, emergency coordinator.
- Systems: HRIS, payroll, electronic medical record, facility registry, national identity system, NHWA platform.
- For each persona: goals, tasks, difficulties, digital skills and connectivity.

Persona **Health worker.** *Person.* A person whose job is to protect and improve the health of their community: holds a registration, licence and employment record in the registry, and needs portable, verifiable credentials and simple renewals.

Specialises the SMART Guidelines generic persona: HealthcareProvider; CommunityHealthWorker.

	Draft for working-group review	Outline basis
Goals	Portable, verifiable credentials; simpler renewals; recognition of training and continuing professional development; able to see and correct their own record.	1.4, 6.3, 12.3
Tasks	Apply for registration and licence; renew with CPD recorded; update employment or place of practice; view and correct own record; present a digital credential for verification online or offline.	5.2, 7.3, 12.3, 20.1, 20.2
Difficulties	Slow licensing; duplicate records across systems; administrative burden of re-entering the same data; concerns about how their data are used.	1.1, 15 (key messages), 14.4, 18.1
Digital skills	Varies. Paper or assisted channels are kept where digital skills are limited. *To confirm with country input.*	15.2

	Draft for working-group review	Outline basis
Connectivity	Varies, including low-connectivity settings; credentials must be verifiable offline. *To confirm with country input.*	7 (key messages), 8.3, 20.2

Persona **Registration officer (regulatory body).** *Person.* Staff of a professional council or regulatory body who registers and licenses practitioners, records status changes and answers verification requests.

	Draft for working-group review	Outline basis
Goals	An accurate register that protects the public by confirming who is qualified and in good standing; faster, cheaper registration, renewal and verification.	1.2, 4.4
Tasks	Register and license practitioners; verify qualifications with training institutions; record status changes (suspended, lapsed, reinstated); answer verification requests, including from other countries; issue certificates of good standing.	7.3, 7.4, 20.1, 20.4
Difficulties	Records on paper or in stand-alone systems; duplicate records; concern about loss of control over their data and workload.	1.1, 13.2, 14.4
Digital skills	*To confirm with country input.*	—
Connectivity	*To confirm with country input.*	—

Persona **Employer or facility human resources manager.** *Person.* Recruits, deploys and manages health workers for a public or private employer or facility; verifies credentials and records changes of employment and place of practice.

Specialises the SMART Guidelines generic persona: HealthSystemManager.

	Draft for working-group review	Outline basis
Goals	Confidence that staff are qualified and authorized to practise; payroll that matches the real workforce.	1.2, 1.4
Tasks	Verify a health worker's status before and during employment; record changes of employment and place of practice; act on notices of suspension.	5.2, 7.3, 20.1, 20.2
Difficulties	Unverified credentials; duplicate and ghost records; data re-entered across HR and payroll systems.	1.1, 9 (key messages)
Digital skills	*To confirm with country input.*	—
Connectivity	*To confirm with country input.*	—

Persona **Training institution registrar.** *Person.* Records enrolment and graduation in a pre-service education institution and confirms qualifications to the regulator.

	Draft for working-group review	Outline basis
Goals	Graduates registered without delay, with their qualifications recognized.	1.4, 20.1
Tasks	Record enrolment and graduation; confirm qualifications to the regulator as the primary source.	4.3, 13.2, 20.1
Difficulties	Education data held in	1.1

	Draft for working-group review	Outline basis
	systems not connected to the regulator's.	
Digital skills	*To confirm with country input.*	—
Connectivity	*To confirm with country input.*	—

Persona **Workforce planner.** *Person.* Uses stock, distribution, skill mix, attrition and migration data for planning and NHWA reporting.

Specialises the SMART Guidelines generic persona: HealthSystemManager; DataManager.

	Draft for working-group review	Outline basis
Goals	Reliable numbers for planning, financing, emergency response and international reporting.	1.4
Tasks	Produce counts and distribution; analyse stock, distribution, skill mix, attrition and migration; prepare NHWA extracts; track inflows and outflows for reporting under the WHO Global Code of Practice.	4.4, 5.2, 7.5, 11.4, 20.4
Difficulties	Weak planning evidence and difficulty reporting to NHWA; incomplete coverage, especially of the private sector.	1.1, 13.1
Digital skills	*To confirm with country input.*	—
Connectivity	*To confirm with country input.*	—

Persona **Member of the public.** *Person.* Checks that a practitioner is qualified and authorized to practise.

Specialises the SMART Guidelines generic persona: Person.

	Draft for working-group review	Outline basis
Goals	Confidence that a practitioner is qualified and authorized to practise.	1.4
Tasks	Check a health worker's status online.	20.2
Difficulties	Sees only what is decided to be public (tiered disclosure).	12.1
Digital skills	*To confirm with country input.*	—
Connectivity	*To confirm with country input.*	—

Persona **Emergency coordinator.** *Person.* Identifies, verifies and deploys qualified health workers, emergency medical teams and volunteers during an emergency.

	Draft for working-group review	Outline basis
Goals	Rapid surge deployment of qualified health workers.	20.3
Tasks	Identify available, qualified workers by skill and location; verify credentials of emergency medical teams and volunteers; track deployment and return.	20.3
Difficulties	Unverified credentials; fragile or insecure settings.	1.1, 21.3
Digital skills	*To confirm with country input.*	—
Connectivity	May work in disrupted settings that need offline options. *To confirm with country input.*	21.3

Persona **Human resources information system (HRIS).** *System.* Manages employment transactions, payroll processing, leave and scheduling; supplies employment data to the registry and uses its identities.

	Draft for working-group review	Outline basis
Purpose	Human resource management: recruitment, appraisal, leave and payroll processing (out of scope for the registry).	2.2
Exchanges with the registry	Supplies employment data to the registry and draws identities and credentials from it.	4 (key messages), 9.1
Issues	Many countries run the registry inside an HRIS; the boundary is a TWG point for discussion.	About this draft
Exchange pattern	Query on demand and notification of changes, or periodic batch. *To confirm with country input.*	9.2, 10.3

Persona **Payroll system.** *System.* Pays health workers; reconciled against the registry to remove duplicate and ghost records.

	Draft for working-group review	Outline basis
Purpose	Pays health workers.	2.2, 9.1
Exchanges with the registry	Reconciled against the registry to find duplicate and ghost records.	1.2, 13.3
Issues	Ghost records.	1.1
Exchange pattern	Periodic reconciliation. *To confirm with country input.*	13.3

Persona **Electronic medical record.** *System.* Identifies and authorizes health workers at the point of care, for example a prescriber, by querying the registry.

Specialises the SMART Guidelines generic persona: System.EMR.

	Draft for working-group review	Outline basis
Purpose	Clinical record at the point of care.	9.1
Exchanges with the registry	Identifies and authorizes health workers, for example checking that a prescriber is authorized.	4.4, 20.2
Issues	*To confirm with country input.*	—
Exchange pattern	Query on demand.	10.2, 10.3

Persona **Facility registry.** *System.* The authoritative list of health facilities; the registry links each place of practice to it.

	Draft for working-group review	Outline basis
Purpose	The authoritative list of health facilities.	9.1
Exchanges with the registry	Place of practice is linked to the facility registry.	9 (key messages), 11.1
Issues	*To confirm with country input.*	—
Exchange pattern	*To confirm with country input.*	—

Persona **National identity system.** *System.* Foundational identity; the registry may link each health worker to a national identity number where available and lawful.

Specialises the SMART Guidelines generic persona: System.ClientRegistry.

	Draft for working-group review	Outline basis
Purpose	Foundational identity and civil registration.	9.1
Exchanges with the registry	The health worker identifier is linked to the national identity number where available and lawful; identity is verified against it.	7.1, 13.2
Issues	Lawful basis for the link.	6.2, 7.1

	Draft for working-group review	Outline basis
Exchange pattern	*To confirm with country input.*	—

Persona **NHWA platform.** *System.* The National Health Workforce Accounts data platform; receives indicator data derived from the registry.

Specialises the SMART Guidelines generic persona: System.HMIS.

	Draft for working-group review	Outline basis
Purpose	National Health Workforce Accounts reporting.	9.1
Exchanges with the registry	Receives indicator extracts derived from registry data.	7.5, 11.4
Issues	Some indicators need other sources, such as finance data and labour force surveys.	11.4
Exchange pattern	Bulk exchange through periodic extracts.	10.3

5.2 Priority use cases

- Register and license a new graduate; renew a licence; update employment or place of practice.
- Verify a health worker's status (by an employer, a patient, another system or another country).
- Produce workforce counts and distribution for planning and NHWA reporting.

User scenario **Register and license a new graduate.** A new graduate applies for registration; the qualification is verified with the training institution; the regulator registers and licenses the graduate; the registry record is created with a unique health worker identifier.

Personas: Health worker, Training institution registrar, Registration officer (regulatory body)

User scenario **Renew a licence.** A health worker renews a time-limited licence, with continuing professional development recorded; the registry updates licence status and validity.

Personas: Health worker, Registration officer (regulatory body)

User scenario **Update employment or place of practice.** An employer or the health worker records a change of employment or place of practice, linked to the facility registry.

Personas: Health worker, Employer or facility human resources manager, Human resources information system (HRIS), Facility registry

User scenario **Verify a health worker's status.** An employer, a patient, another system or another country checks a health worker's registration and licence status.

Personas: Employer or facility human resources manager, Member of the public, Electronic medical record, Registration officer (regulatory body)

User scenario **Produce workforce counts and distribution for planning and NHWA reporting.** Workforce counts and distribution are produced from the registry for national planning and for reporting to the National Health Workforce Accounts.

Personas: Workforce planner, NHWA platform

5.3 User stories

- Standard format: "As a [persona], I need [capability], so that [benefit]."
- Worked examples for each priority use case.

5.4 Requirements derivation

- The chain from use case to business process, to functional and non-functional requirements, to data elements.
- Traceability between each requirement and the use case it serves.
- Prioritization (must have, should have, could have) to support phased delivery.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: persona cards.
- Table: use case overview (actor, trigger, data used, systems involved).
- Figure: traceability from use case to requirement to data element.

Implementation tools and checklists

- Persona template (full generic personas in annex).
- Use case prioritization matrix (value against feasibility).
- User story template.

Expected outputs for countries

- National personas adapted from the generic set.
- A prioritized list of use cases for the first phase.
- An initial set of user stories.

Chapter 6. Governance and Stewardship

Chapter objectives. This chapter will help readers to:

- Describe the institutional arrangements needed to govern a registry across health workforce, regulatory and digital bodies.
- Identify the legal and policy foundations for collecting, sharing and publishing health worker data.
- Plan for long-term stewardship, financing and sustainability.

Key messages

- Weak governance, not technology, is the most common reason registries stall or fail.
- The registry needs a clear owner, a mandate, and formal agreements with every institution that supplies data.
- Health workers' personal data must be protected in law and in practice, and health workers should be able to see and correct their own record.
- Operating costs and staffing must be budgeted from the start, beyond project funding.

6.1 Institutions

- Roles of the ministry of health (health workforce and digital units), regulatory councils, public service commission, finance and payroll, education ministry and training institutions, private sector and sub-national authorities.
- Governance structure: steering committee, technical working group, and registry operator or custodian.
- Decision rights: who approves data standards, access, changes and new connections.

6.2 Legal foundations

- A legal mandate to establish and operate the registry and to require institutions to submit data.
- Data protection, privacy and consent, including what may be public (for example licence status) and what is restricted.
- Data-sharing agreements between institutions and, where relevant, across borders.

6.3 Data stewardship

- Named data owners, stewards and custodians for each group of data.
- Policies for data access, use, retention, correction and release.
- Health workers' rights to access and correct their own record.

6.4 Sustainability

- Options for recurrent financing: government budget, registration and licensing fees, shared digital infrastructure budgets.
- Institutional capacity: permanent posts for registry operation, data management and technical support.
- Managing dependence on suppliers and external partners, including handover and exit planning.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: model governance structure.
- Table: responsibility matrix (who is responsible, accountable, consulted and informed) for registry functions.
- Table: legal and policy instruments typically needed.

Implementation tools and checklists

- Governance terms of reference template (annex).
- Data-sharing agreement template.
- Legal and policy checklist.

Expected outputs for countries

- A designated registry owner and governance body.
- Signed data-sharing agreements with data-supplying institutions.
- A costed sustainability plan.

Chapter 7. Functional Requirements

Chapter objectives. This chapter will help readers to:

- Describe the core functions that any health workforce registry should provide.
- Distinguish minimum functions from optional or advanced ones.
- Provide a requirements catalogue that countries can adapt for specification and procurement.

Key messages

- The core is small: identify each worker uniquely, hold a trusted record, keep it current, and answer queries.
- Functions can be delivered by one system or shared across several connected systems.
- Non-functional requirements (security, availability, offline use, languages) matter as much as functions.
- Requirements should be written so that they can be tested.

7.1 Identity management

- Assign a unique, persistent health worker identifier, linked to the national identity number where available and lawful.
- Search, match and merge duplicate records across source systems.
- Manage changes of name and other identity details over a career.

Functional requirement FR-ID-01 Assign a unique, persistent health worker identifier, linked to the national identity number where available and lawful.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.1 Identity management, word for word. The identifier FR-ID-01 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-006-02; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Functional requirement FR-ID-02 Search, match and merge duplicate records across source systems.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.1 Identity management, word for word. The identifier FR-ID-02 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-031-01; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Functional requirement FR-ID-03 Manage changes of name and other identity details over a career.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.1 Identity management, word for word. The identifier FR-ID-03 was assigned by an agent and is not in the source.

7.2 Records

- Create, view, update and archive the health worker record in line with the minimum data set.
- Hold supporting evidence, such as certificates, or links to it.
- Record the source and date of each data item.

Functional requirement FR-REC-01 Create, view, update and archive the health worker record in line with the minimum data set.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.2 Records, word for word. The identifier FR-REC-01 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-006-01, fr-sec-006-05; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Functional requirement FR-REC-02 Hold supporting evidence, such as certificates, or links to it.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.2 Records, word for word. The identifier FR-REC-02 was assigned by an agent and is not in the source.

Functional requirement FR-REC-03 Record the source and date of each data item.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.2 Records, word for word. The identifier FR-REC-03 was assigned by an agent and is not in the source.

7.3 Lifecycle management

- Manage status changes: registered, licensed, suspended, lapsed, retired, deceased, emigrated.
- Support renewals, recording of continuing professional development, and reminders.
- Record changes of employment and place of practice over time.

Functional requirement FR-LC-01 Manage status changes: registered, licensed, suspended, lapsed, retired, deceased, emigrated.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.3 Lifecycle management, word for word. The identifier FR-LC-01 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-009-05; system type(s) hrh-information-system. Agent-proposed, for working-group review.

Functional requirement FR-LC-02 Support renewals, recording of continuing professional development, and reminders.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.3 Lifecycle management, word for word. The identifier FR-LC-02 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-009-05; system type(s) hrh-information-system. Agent-proposed, for working-group review.

Functional requirement FR-LC-03 Record changes of employment and place of practice over time.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.3 Lifecycle management, word for word. The identifier FR-LC-03 was assigned by an agent and is not in the source.

7.4 Verification services

- Confirm registration and licence status in real time for employers, the public and other systems.

- Issue digital, verifiable credentials or certificates (for example a certificate of good standing).
- Respond to verification requests from other countries.

Functional requirement FR-VER-01 Confirm registration and licence status in real time for employers, the public and other systems.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.4 Verification services, word for word. The identifier FR-VER-01 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-006-02, fr-sec-031-02; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Functional requirement FR-VER-02 Issue digital, verifiable credentials or certificates (for example a certificate of good standing).

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.4 Verification services, word for word. The identifier FR-VER-02 was assigned by an agent and is not in the source.

Functional requirement FR-VER-03 Respond to verification requests from other countries.

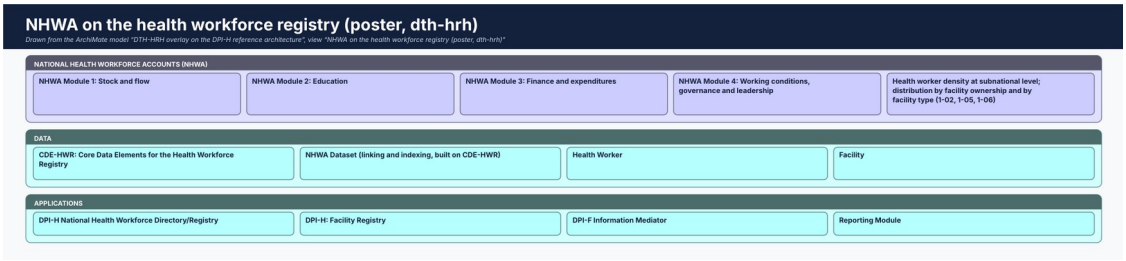
Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.4 Verification services, word for word. The identifier FR-VER-03 was assigned by an agent and is not in the source.

7.5 Analytics

- Standard reports and dashboards on stock, distribution, age, sex, occupation and status.
- Data extracts for NHWA and national planning.
- Data quality reports for registry managers.

Figure.

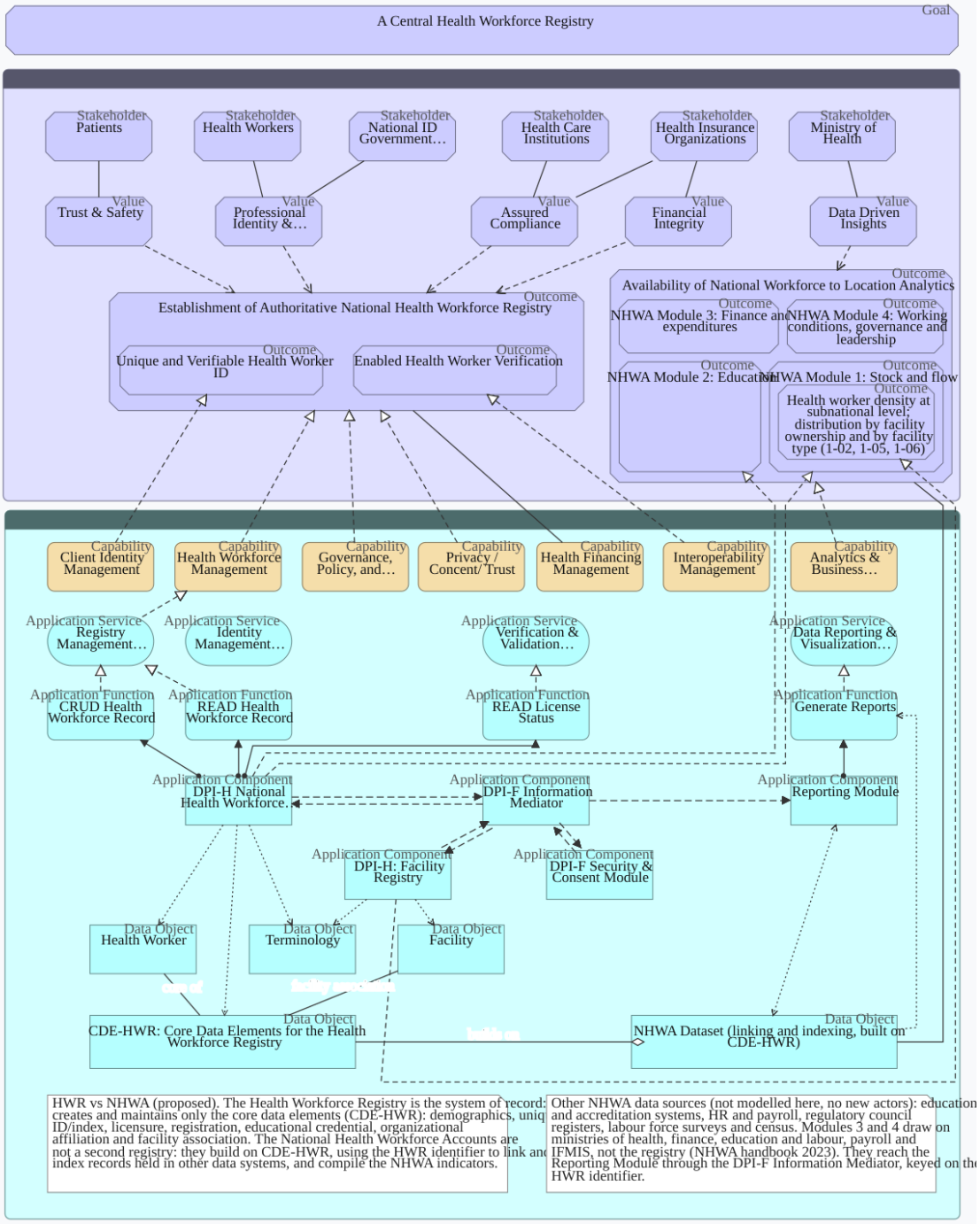


Source: the dth-hrh ArchiMate overlay 0.1.0-draft.1 on the smart-ra (DPI-H RA) model 0.2.0, view "NHWA on the health workforce registry (poster, dth-hrh)", drawn from the model by the `archimate-blocks` tool. The NHWA elements belong to this handbook's overlay, not to smart-ra. Agent-proposed, for working-group review (dth-hrh#47).

National Health Workforce Accounts (NHWA) on the health workforce registry: the four NHWA modules and the subnational indicators, the data they draw on, and the applications that hold and move it

Figure.

Health workforce registry and NHWA (dth-hrh)
Drawn from the ArchiMate model "DTH-HRH overlay on the DPI-H reference architecture"



Key

Colour: which part of the architecture a box belongs to

- Why: who cares (stakeholders), and the drivers, goals, outcomes, values and requirements
- Strategy: what the organisation is able to do (capabilities) and with what (resources, value streams)
- Software: systems and their parts, the services they offer, and the data they hold

Shape: what kind of thing a box is (its small italic label names the exact kind)

- Cut corners: a reason or aim, such as a stakeholder, goal, outcome or value
- Rounded corners: something done or an ability, such as a capability, process or function
- Rounded ends: a service, offered for others to use
- Square corners: something that exists, such as a system, a role, or a piece of data

Other marks

Source: the dth-hrh ArchiMate overlay 0.1.0-draft.1 on the smart-ra (DPI-H RA) model 0.2.0, view "Health workforce registry and NHWA (dth-hrh)", a copy of smart-ra view "1. Central Health Workforce Registry" with the NHWA elements added, drawn from the model by the `archimate-blocks` tool. The NHWA elements belong to this handbook's overlay, not to smart-ra. Agent-proposed, for working-group review (dth-hrh#47).

The health workforce registry and the National Health Workforce Accounts (NHWA): the registry view of the reference architecture with the NHWA work this handbook adds

Functional requirement FR-AN-01 Standard reports and dashboards on stock, distribution, age, sex, occupation and status.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.5 Analytics, word for word. The identifier FR-AN-01 was assigned by an agent and is not in the source.

NHWA indicators: 1-01, 1-02, 1-03, 1-04, 1-05, 1-06, from *National health workforce accounts: a handbook*, 2nd edition (WHO, 2023), where they name a health workforce registry as a data source. Agent-proposed link, for working-group review.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-006-04, fr-sec-037-02, fr-sec-038-05; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Functional requirement FR-AN-02 Data extracts for NHWA and national planning.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.5 Analytics, word for word. The identifier FR-AN-02 was assigned by an agent and is not in the source.

NHWA indicators: 1-01, 1-02, 1-03, 1-04, 1-05, 1-06, 1-07, 1-08, 1-09, 1-10, 1-12, 2-04, from *National health workforce accounts: a handbook*, 2nd edition (WHO, 2023), where they name a health workforce registry as a data source. Agent-proposed link, for working-group review.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-006-04; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Functional requirement FR-AN-03 Data quality reports for registry managers.

Priority: to be set (must / should / could). *Linked use cases:* to be traced.

Source: the TWG outline (L1, twg-outline-2026-10-02), section 7.5 Analytics, word for word. The identifier FR-AN-03 was assigned by an agent and is not in the source.

External glossary reference (not definitional): who-iris glossary `hrhis-registry-2015`, concept(s) fr-sec-006-03, fr-sec-037-09; system type(s) health-workforce-registry. Agent-proposed, for working-group review.

Non-functional requirements

Non-functional requirements matter as much as functions. The initial set, from the chapter's key messages:

Non-functional requirement NFR-01 **Security.** Health workers' personal data are protected in storage and in transit, with role-based access for people and authenticated access for systems (see Chapter 12).

Non-functional requirement NFR-02 **Availability.** Verification services are available when employers, the public and other systems need them, with agreed service levels.

Non-functional requirement NFR-03 **Offline and low-connectivity use.** Registration and verification work in low-connectivity settings, including through paper or assisted channels where needed.

Non-functional requirement NFR-04 **Languages.** The registry and its public views support the languages its users need.

Non-functional requirement NFR-05 **Testable requirements.** Every requirement is written so that conformance to it can be tested.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: overview of registry functions.
- Table: minimum and advanced functions.
- Table: sample requirements (identifier, requirement, priority, linked use case).

Implementation tools and checklists

- Functional requirements catalogue (annex).
- Non-functional requirements checklist.

Expected outputs for countries

- A national requirements specification, adapted from the catalogue and prioritized by phase.

Chapter 8. Digital Architecture Principles

Chapter objectives. This chapter will help readers to:

- Explain the main architecture options for a registry in non-technical terms.
- Help countries choose a pattern that fits their governance, existing systems and capacity.
- Set out principles that keep the registry scalable, reusable and independent of any one product.

Key messages

- There is no single correct architecture. The choice depends on how authority over health worker data is distributed in the country.
- Centralized, federated and hybrid models can all provide one consistent view of the workforce.
- Architecture should follow governance: do not centralize technically what is decentralized legally.
- Open standards and interfaces protect the investment whichever product is used.

8.1 Architecture patterns

- The three main patterns and what each means in practice: centralized (one national system holds the master record), federated (source systems keep their data and are linked), and hybrid (a national index with detail held at source).
- Criteria for choosing: legal mandates, number of regulators, existing systems, connectivity and capacity.
- Fit with the national digital health enterprise architecture and the registry layer of the draft DPI-H reference architecture.

8.2 Federation

- How several regulators or sub-national registries contribute to a national view through a shared identifier, a common data set and common interfaces.
- Linking records and resolving conflicts between sources.
- Responsibilities that stay with each source and those that sit at national level.

8.3 Scalability

- Start with priority occupations or regions and expand without redesign.
- Performance, availability and hosting options (government data centre or cloud), including for low-connectivity settings.
- Modular design, so that functions can be added or replaced independently.

8.4 Technology neutrality

- Specify standards and interfaces, not products.
- Considerations for open-source, commercial and custom-built options, including digital public goods.
- Avoiding dependence on one supplier: data ownership, data export, documentation and exit clauses.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: the three architecture patterns side by side.
- Figure: decision tree for choosing a pattern.
- Table: comparison of patterns (advantages, risks, prerequisites).

Implementation tools and checklists

- Architecture pattern selection checklist.
- Architecture principles template for national adoption.

Expected outputs for countries

- A chosen architecture pattern with the reasons for the choice.
- An adopted set of national architecture principles for the registry.

Chapter 9. Registry within Ecosystem

Part 2. Technical Design. This part gives the technical guidance needed to design a registry that connects with the wider health and government digital ecosystem: where the registry sits, how it exchanges data, what data it holds, and how those data are kept secure and of good quality.

Chapter objectives. This chapter will help readers to:

- Map the systems that supply data to the registry or use data from it.
- Describe the main data flows between these systems.
- Illustrate how the registry fits into different national architectures.

Key messages

- The registry is one of several foundational registries. It is most valuable when linked to the facility registry and to identity systems.
- Every connection needs a defined purpose, an owner and a documented data flow.
- Mapping what already exists is the first design step.
- Data should be captured once at source and reused, not re-entered.

9.1 System landscape

- Source systems: regulatory council systems, training institution records, HRIS, payroll, national identity and civil registration.
- Systems that use registry data: electronic medical records, electronic prescribing, telehealth, insurance and claims, health management information systems, emergency rosters, the NHWA platform.
- Shared services: facility registry, terminology service, interoperability layer, authentication.

9.2 Data flows

- Who sends what to whom, when, and which event triggers it.
- Flows into the registry (new records and updates) and out of it (queries, notifications, reports).
- Periodic batch exchange compared with real-time exchange.

9.3 Architecture examples

- Illustrative architectures for centralized, federated and hybrid models.
- An example in which the registry is part of a national health information exchange.
- An example for a country starting from paper registers or spreadsheets.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: ecosystem landscape diagram.
- Figure: data flow diagram.
- Table: systems, their owners, and the data they supply or use.

Implementation tools and checklists

- System inventory template.
- Data flow mapping template.

Expected outputs for countries

- A national map of registry-related systems.
- Documented current and target data flows.

Chapter 10. Interoperability

Chapter objectives. This chapter will help readers to:

- Explain the standards and interfaces that allow the registry to exchange data with other systems.
- Describe common data exchange patterns and when to use each.
- Define how interoperability is governed and tested.

Key messages

- Interoperability means that systems can exchange information and use it without manual re-entry.
- Use existing open standards. Do not create national formats where a standard already exists.
- Agreeing on meaning (shared codes and definitions) is as important as agreeing on technical format.
- Interoperability needs ongoing governance: rules for who can connect, and testing that they conform.

10.1 Standards

- HL7 FHIR resources for health workers and where they work (Practitioner, PractitionerRole, Organization, Location) as the recommended exchange format.

- IHE profiles such as Mobile Care Services Discovery (mCSD) for querying and updating health worker and facility directories.
- Code systems and classifications, covered in the chapter on data standards.

10.2 APIs

- Core interfaces: search for and read a health worker record; verify status; submit and update records; receive notice of changes.
- Security of interfaces: authentication, authorization and consent.
- Documentation and version management, so that other systems can connect.

10.3 Data exchange patterns

- Query on demand, notification of changes, and bulk exchange through periodic files or extracts.
- Direct connection compared with exchange through an interoperability layer.
- Transitional approaches where source systems cannot yet use standards (for example structured file uploads).

10.4 Governance

- Rules and agreements for systems that connect to the registry.
- Conformance testing against published specifications before a connection goes live.
- Managing changes to standards and interface versions.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: layers of interoperability (legal, organizational, semantic, technical).
- Table: standards and what each is used for.
- Figure: step-by-step sequence of a verification query.

Implementation tools and checklists

- Interoperability models (annex).
- Interface specification checklist.
- Conformance test checklist.

Expected outputs for countries

- A national interoperability specification for the registry.
- A procedure for connecting new systems.

Chapter 11. Data Standards and Minimum Dataset

Chapter objectives. This chapter will help readers to:

- Present a minimum data set for a health workforce registry and the reason for each element.
- Identify the classifications and code lists that make data comparable.
- Show how registry data map to NHWA indicators.

Key messages

- Collect only what is needed, define it precisely, and keep it up to date.
- The minimum data set is a floor, not a ceiling. Countries can extend it.
- International classifications allow comparison across institutions and countries.
- A well-designed registry can produce many NHWA indicators as a by-product of routine operation.

11.1 Core data elements (CDE-HRH)

- Core data elements (CDE-HRH), held by every registry, each with a working definition for working-group review: identification number, name, birth history, sex, citizenship and language, address, contact information, occupation, professional licence and certification, employment status, employment address, and data submission institution.
- Sex and occupation are core because NHWA indicators are broken down by occupation, and many by sex.
- Sections 11.2 to 11.5 add extension data elements for each NHWA module (National health workforce accounts handbook, 2nd edition).
- Updates to consider: unique health worker identifier, more detail on education and qualifications, specialty, licence status and validity, place of practice linked to the facility registry, sector, and continuing professional development.
- For each element: definition, format, whether mandatory, source and sensitivity.
- Terms and definitions follow the TWG glossary (draft, 2026-09-29), the external source this handbook uses for agreed terms.

Table.

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
1	Identification number	—	—	To draft
2	Name	—	—	To draft
3	Birth history	1-03, 1-07	—	To draft
4	Sex	1-03, 1-04, 1-09, 1-10, 2-04	—	Provisional (NHWA)
5	Citizenship and language	1-07	—	To draft
6	Address	—	—	To draft
7	Contact	—	—	To draft

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
	information			
8	Occupation	1-01, 1-02, 1-03, 1-04, 1-05, 1-06, 1-07, 1-08, 1-09, 1-10, 1-12, 3-02	—	Provisional (NHWA)
9	Professional licence and certification	1-01	—	To draft
10	Employment status	1-01, 1-09, 1-10	—	To draft
11	Employment address	1-02, 1-05, 1-06	—	To draft
12	Data submission institution	—	—	To draft

787 *Core data elements (CDE-HRH), at a glance*

788 Data element **Identification number**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

789 Data element **Name**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

790 Data element **Birth history**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

791 Data element **Sex**: working definition, for working-group review.

792 NHWA indicators: 1-03, 1-04, 1-09, 1-10, 2-04.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Citizenship and language**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Address**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Contact information**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Occupation**: working definition, for working-group review.

NHWA indicators: 1-01, 1-02, 1-03, 1-04, 1-05, 1-06, 1-07, 1-08, 1-09, 1-10, 1-12, 3-02.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Professional licence and certification**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Employment status**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Employment address**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Data submission institution**: working definition, for working-group review.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

11.2 NHWA Module 1: Stock and flow

- Extension data elements a registry needs to report NHWA Module 1 (stock and flow), on top of the core data elements.
- Indicators 1-01 to 1-10 and 1-12 name a health workforce registry or database as a data source.
- Indicator 1-11 (vacancy rate) counts posts, not health workers, and comes from other sources.

Table.

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
1	Activity level	1-01	practising, professionally active, licensed to practise	Provisional (NHWA)
2	Subnational unit of practice	1-02	level 1 subnational administrative units	Provisional (NHWA)
3	Facility ownership	1-05	public, private not-for-profit, private for-profit	Provisional (NHWA)
4	Facility type	1-06	—	Provisional (NHWA)
5	Place of training	1-08, 1-09	—	Provisional (NHWA)
6	Date of entry into the active	1-09	—	To draft

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
	workforce			
7	Date and type of exit from the active workforce	1-10	voluntary, involuntary	To draft
8	Type of contract	1-12	—	Provisional (NHWA)

NHWA Module 1 (stock and flow) extension data elements, at a glance

Data element **Activity level**: working definition, for working-group review.

NHWA indicators: 1-01. Categories named in the NHWA handbook: practising, professionally active, licensed to practise.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Subnational unit of practice**: working definition, for working-group review.

NHWA indicators: 1-02. Categories named in the NHWA handbook: level 1 subnational administrative units.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Facility ownership**: working definition, for working-group review.

NHWA indicators: 1-05. Categories named in the NHWA handbook: public, private not-for-profit, private for-profit.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Facility type**: working definition, for working-group review.

NHWA indicators: 1-06.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Place of training**: working definition, for working-group review.

NHWA indicators: 1-08, 1-09.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Date of entry into the active workforce**: working definition, for working-group review.

NHWA indicators: 1-09.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Date and type of exit from the active workforce**: working definition, for working-group review.

NHWA indicators: 1-10. Categories named in the NHWA handbook: voluntary, involuntary.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Type of contract**: working definition, for working-group review.

NHWA indicators: 1-12.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

11.3 NHWA Module 2: Education

- Extension data elements a registry needs to report NHWA Module 2 (education).
- Of the module's indicators, 2-04 (ratio of graduates to stock) names a health workforce registry or database as a data source.
- Indicators 2-01 to 2-03 and 2-05 to 2-07 describe education places, applications and programmes, and come from ministries of education and training institutions.

Table.

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
1	Qualification and	2-04	—	To draft

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
	education programme			
2	Training institution and year of graduation	2-04	—	To draft

NHWA Module 2 (education) extension data elements, at a glance

Data element **Qualification and education programme**: working definition, for working-group review.

NHWA indicators: 2-04.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Training institution and year of graduation**: working definition, for working-group review.

NHWA indicators: 2-04.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

11.4 NHWA Module 3: Finance and expenditures

- Extension data elements a registry can hold to support NHWA Module 3 (finance and expenditures). They are optional.
- The module's indicators are mostly expenditure totals from finance and health ministry records, not registry data.

Table.

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
1	Employer sector	3-01, 3-02	public, private	Provisional (NHWA)
2	Salary grade at entry	3-02	—	To draft

NHWA Module 3 (finance and expenditures) extension data elements, at a glance

Data element **Employer sector**: working definition, for working-group review.

NHWA indicators: 3-01, 3-02. Categories named in the NHWA handbook: public, private.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

Data element **Salary grade at entry**: working definition, for working-group review.

NHWA indicators: 3-02.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

11.5 NHWA Module 4: Working conditions, governance and leadership

- Extension data elements a registry can hold to support NHWA Module 4 (working conditions, governance and leadership). They are optional.
- Indicators 4-01, 4-02 and 4-05 describe policies and national capacity, not individual health workers.

Table.

#	Data element	NHWA indicators	Categories (NHWA handbook)	Definition
1	Leadership position	4-03	—	Provisional (NHWA)

NHWA Module 4 (working conditions, governance and leadership) extension data elements, at a glance

Data element **Leadership position**: working definition, for working-group review.

NHWA indicators: 4-03.

Definition	Format	Mandatory	Source	Sensitivity	FHIR mapping
to draft	*to draft*	*to draft*	*to draft*	*to draft*	*to draft*

11.6 Classifications

- Occupations: International Standard Classification of Occupations (ISCO-08), with mapping from national occupational titles.
- Education levels and fields (ISCED), facility types and ownership, and administrative geography.
- Supporting standards for countries (ISO 3166), dates (ISO 8601) and languages.

11.7 Metadata

- Data dictionary structure: element name, definition, data type, permitted values and mapping to FHIR.
- Information about each data item's origin: source, date recorded, date verified.
- Version control and publication of the data dictionary and code lists.

11.8 NHWA mapping

- NHWA indicators that can be derived from registry data, such as stock, density, distribution, age and sex, and place of training.
- Indicators that need other sources, such as finance data and labour market surveys.
- The breakdowns that NHWA requires and what they mean for the data set.

Figures, tools and expected outputs

Suggested figures and tables

- Table: minimum data set.
- Figure: logical data model (health worker, qualification, licence, role, facility).
- Table: registry data elements mapped to NHWA indicators.

Implementation tools and checklists

- Minimum data set (annex) and data dictionary (annex).
- Template for mapping national occupational titles to ISCO-08.
- NHWA mapping (annex).

Expected outputs for countries

- A national minimum data set and data dictionary.
- National occupational titles mapped to ISCO-08.
- A list of the NHWA indicators that the registry will supply.

Chapter 12. Privacy, Security and Trust

Chapter objectives. This chapter will help readers to:

- Apply privacy by design to health workers' personal data.
- Identify the security controls that a registry needs.
- Describe how access is controlled and audited so that stakeholders can trust the registry.

Key messages

- A registry holds personal data about health workers. Protecting those data is a legal and ethical duty.
- Decide what is public, what is shared between institutions and what is restricted, and be open about it.

- Security is a continuous activity, not a one-off installation.
- Trust depends on transparency: health workers should know who uses their data and why.

12.1 Privacy by design

- Collect the minimum data needed, use them only for stated purposes, and set retention limits.
- Carry out a privacy impact assessment before implementation and before major changes.
- Tiered disclosure: a public view, an institutional view and restricted data.

12.2 Cybersecurity

- Risk assessment, secure hosting, encryption of stored and transmitted data, backup and recovery.
- Incident response and notification of data breaches.
- Alignment with national cybersecurity policies and standards.

12.3 Access control

- Role-based access for people; authenticated and authorized access for systems.
- Self-service access for health workers to their own record.
- Regular review of user accounts and permissions.

12.4 Audit trails

- A log of who viewed or changed which record, when and why.
- Routine review of logs, with reports to the governance body.
- Use of audit data to detect misuse and fraud.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: tiered data access model.
- Table: main threats and the controls that address them.
- Table: roles and permissions.

Implementation tools and checklists

- Privacy impact assessment template.
- Security checklist.
- Roles and permissions template.

Expected outputs for countries

- A completed privacy impact assessment.
- A data classification and disclosure policy.
- A security plan.

Chapter 13. Data Quality Management

Chapter objectives. This chapter will help readers to:

- Define data quality for a registry and how to measure it.
- Describe validation when data are entered and when they are compared across sources.
- Set up routine monitoring and improvement of data quality.

Key messages

- A registry is only as useful as its data are complete, accurate and current.
- Quality is best assured at source, by the institution that owns the data.
- Routine use of the data is the strongest driver of quality.
- Data quality needs named responsibility and regular reporting.

13.1 Quality dimensions

- Completeness, accuracy, timeliness, uniqueness, consistency and validity.
- Coverage: the proportion of the actual workforce that is in the registry, including the private sector.
- Targets for each dimension.

13.2 Validation

- Automated rules at data entry (formats, mandatory fields, code lists).
- Verification against primary sources, such as training institutions, the national identity system and the issuing regulator.
- Procedures for finding and resolving duplicate records.

13.3 Monitoring

- Data quality indicators and dashboards.
- Periodic audits and reconciliation with payroll, facility staffing lists or census data.
- Feedback to data suppliers and corrective action.

Figures, tools and expected outputs

Suggested figures and tables

- Table: data quality dimensions with example indicators.
- Figure: data quality improvement cycle.

Implementation tools and checklists

- Data quality assessment checklist.
- Catalogue of validation rules.
- Standard operating procedure template for resolving duplicates.

Expected outputs for countries

- A data quality plan with indicators and targets.
- A routine data quality report.

Chapter 14. Preparing for Transformation

Part 3. Country Implementation. This part guides countries through the practical steps of transformation, from preparing and assessing readiness to operating and improving the registry. Countries can enter at the step that matches their situation.

Chapter objectives. This chapter will help readers to:

- Secure leadership and form the team that will lead the transformation.
- Assess readiness and the current state of registry-related systems.
- Engage the stakeholders on whose participation the registry depends.

Key messages

- Joint leadership by the health workforce and digital health units is essential.
- A readiness assessment prevents costly surprises later.
- Regulators and professional councils are partners and data owners, not only data suppliers.
- Agree the vision and first-phase scope before choosing technology.

14.1 Leadership

- An executive sponsor and a clear mandate.
- A shared vision and goals linked to the national health workforce and digital health strategies.
- The registry's place in the national digital health architecture and investment plans.

14.2 Teams

- Core team: health workforce lead, digital health lead, regulator representatives, business analyst, architect, data manager and legal adviser.
- Roles, responsibilities and time commitment.
- When to bring in external technical support.

14.3 Readiness

- Assessment areas: governance and law, data, processes, technology and infrastructure, people and skills, and financing.
- An inventory of existing registers and systems.
- Placement on the maturity model and identification of gaps.

14.4 Stakeholder engagement

- Stakeholder mapping, with analysis of interests and influence.
- Engagement methods: workshops, one-to-one consultation with councils, and dialogue with health worker associations.

- Responding to concerns about loss of control, workload and use of data.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: team structure.
- Table: readiness assessment areas and sample questions.
- Figure: stakeholder map.

Implementation tools and checklists

- Readiness assessment (annex).
- Stakeholder mapping template.
- Team terms of reference template.

Expected outputs for countries

- A mandated team and a vision statement.
- A readiness assessment report.
- A stakeholder engagement plan.

Chapter 15. Business Process Redesign

Chapter objectives. This chapter will help readers to:

- Document how registration, licensing and related processes work today.
- Design improved processes that make use of digital capabilities.
- Derive system requirements from the redesigned processes.

Key messages

- Digitizing a poor process produces a poor digital process. Simplify first.
- Mapping processes with the people who do the work reveals the real bottlenecks.
- Redesign should reduce the burden on health workers.
- Some process changes need regulatory or policy change. Identify these early.

15.1 Current state mapping

- Map priority processes: first registration, licence renewal, verification, transfer and exit.
- Record the steps, people involved, documents, time taken and difficulties.
- Identify duplication across institutions.

15.2 Future workflows

- Redesigned workflows using online application, automated checks, shared data and digital credentials.
- Paper or assisted channels kept where connectivity or digital skills are limited.
- Policy or legal changes needed to make the new workflows possible.

15.3 Requirements

- Turn future workflows into functional and non-functional requirements (linked to the chapters on use cases and functional requirements).
- Data requirements for each step.
- Prioritization and phasing.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: example current and future workflow for licence renewal.
- Table: common difficulties and redesign options.

Implementation tools and checklists

- Process mapping template.
- Requirements template.

Expected outputs for countries

- Current and future process maps.
- Validated requirements.
- A list of policy changes needed.

Chapter 16. Planning and Procurement

Chapter objectives. This chapter will help readers to:

- Compare the options for delivering the registry.
- Choose a procurement approach and prepare specifications.
- Develop a costed implementation plan and manage risk.

Key messages

- Consider adapting or connecting existing systems before buying or building new ones.
- The total cost of ownership over at least five years matters more than the purchase price.
- Contracts must protect the country's ownership of its data and avoid dependence on one supplier.
- Plan in phases with clear milestones.

16.1 Options analysis

- Options: enhance an existing system, adopt an open-source solution or digital public good, buy a commercial product, build a custom system, or use a shared government platform.
- Criteria: fit with requirements, conformance to standards, cost, local capacity and sustainability.

- Total cost of ownership and budgeting, following the DIIG approach.

16.2 Procurement approaches

- Specifications and evaluation criteria based on the requirements and standards.
- Contract terms: data ownership, access to source code, support, service levels, training and exit.
- Coordination of funders and alignment with national procurement rules.

16.3 Risk management

- Common risks: weak mandate, stakeholder resistance, poor data, funding gaps, supplier dependence and staff turnover.
- A risk register with mitigation measures.
- Phasing and pilots to reduce risk.

Figures, tools and expected outputs

Suggested figures and tables

- Table: comparison of delivery options.
- Figure: sample phased roadmap.
- Table: risk matrix.

Implementation tools and checklists

- Options analysis template.
- Total cost of ownership and budget template.
- Risk register template.
- Procurement specification checklist.

Expected outputs for countries

- An options appraisal and decision.
- A costed implementation plan and roadmap.
- A risk register.

Chapter 17. Implementation and Deployment

Chapter objectives. This chapter will help readers to:

- Plan the migration of existing records into the registry.
- Test the registry before and during roll-out.
- Roll out in phases and establish routine operations.

Key messages

- Data migration and cleaning usually take longer than expected. Plan and resource them.
- Test with real users and realistic data.

- Roll out in phases, learn, and adjust.
- Put operations and support in place before going live.

17.1 Migration

- Take stock of existing paper and digital records, then clean, de-duplicate and map them.
- Digitize paper registers and verify the records.
- Load data in a planned order, for example by occupation or sector.

17.2 Testing

- Functional, integration, performance, security and user acceptance testing.
- Conformance testing of interfaces.
- Test data and test environments.

17.3 Rollout

- A pilot, followed by phased roll-out by occupation, institution or region.
- Criteria for going live, and a period of running old and new processes in parallel.
- User support during the transition.

17.4 Operations

- Standard operating procedures, help desk and service levels.
- Hosting, backup, maintenance and updates.
- Bringing new data suppliers and connecting systems on board.

Figures, tools and expected outputs

Suggested figures and tables

- Figure: data migration steps.
- Table: types of testing and their purpose.
- Figure: phased roll-out timeline.

Implementation tools and checklists

- Migration plan template.
- Test plan and user acceptance test templates.
- Go-live checklist.
- Standard operating procedure template.

Expected outputs for countries

- Migrated and validated data.
- Test reports.
- An operating registry with procedures and support arrangements.

Chapter 18. Change Management

Chapter objectives. This chapter will help readers to:

- Plan communications for each stakeholder group.
- Promote adoption and routine use of the registry.
- Build the skills needed to operate, use and sustain the registry.

Key messages

- People adopt systems that make their work easier and whose purpose they understand.
- Communicate early, often and in the language of each audience.
- Health workers must see benefits for themselves.
- Capacity strengthening is continuous and should be built into institutions.

18.1 Communications

- Key messages for each audience: health workers, regulators, employers, managers and the public.
- Channels and timing.
- Responding to concerns about data use and privacy.

18.2 Adoption

- Champions, incentives, and links between the registry and processes that users already need, such as licence renewal and employment.
- Monitoring uptake and removing barriers.
- Ways for users to give feedback.

18.3 Capacity strengthening

- Competencies for registry managers, data officers, technical staff and other users.
- Training approaches: on-the-job training, e-learning, and peer learning between councils and countries.
- Building registry skills into pre-service and in-service programmes, and retaining skilled staff.

Figures, tools and expected outputs

Suggested figures and tables

- Table: communication matrix by stakeholder group.
- Table: competencies by role.

Implementation tools and checklists

- Communications plan template.
- Training plan template.
- Change readiness checklist.

Expected outputs for countries

- A communications plan.
- A training plan and materials.
- An adoption monitoring report.

Chapter 19. Monitoring and Continuous Improvement

Chapter objectives. This chapter will help readers to:

- Define indicators to track the registry's performance and use.
- Use a maturity model to plan progress.
- Evaluate results and feed lessons into improvement.

Key messages

- Measure whether the registry is used and useful, not only whether it is installed.
- Maturity is gradual. Set realistic targets for the next level.
- Regular review keeps the registry in step with changing needs.
- Share lessons with other countries.

19.1 KPIs

- Coverage and completeness, timeliness of updates, and data quality.
- Service performance: time to register, renew and verify; system availability.
- Use: number of connected systems, number of verification queries, and reports used for decisions.

19.2 Maturity

- Maturity levels across governance, data, processes, technology, interoperability and use.
- Self-assessment and target-setting.
- Links to existing WHO assessments of health workforce information systems and digital health maturity.

19.3 Evaluation

- Periodic evaluation of results against objectives and costs.
- Learning reviews with stakeholders.
- Updates to the roadmap, requirements and standards.

Figures, tools and expected outputs

Suggested figures and tables

- Table: key performance indicators with definitions and data sources.
- Figure: maturity model summary.
- Figure: results chain from inputs to outcomes.

Implementation tools and checklists

- Indicator framework template.
- Maturity self-assessment (annex).

Expected outputs for countries

- A monitoring and evaluation plan.
- A baseline maturity assessment, repeated periodically.
- An improvement plan.

Chapter 20. Illustrative Use Cases

Part 4. Examples and Resources. This part illustrates the guidance with worked use cases and country experience, and provides practical tools and templates that countries can adapt.

Chapter objectives. This chapter will help readers to:

- Show how the registry supports specific real-world scenarios from start to finish.
- Set out the people, data, workflows and interfaces involved in each scenario.
- Help countries adapt the use cases to their own context.

Key messages

- Use cases make the value of a registry concrete for decision-makers and guide developers.
- The same core data and interfaces serve all four use cases.
- Each use case can be implemented in stages.
- The use cases are illustrative, not prescriptive.

20.1 Licensing

User scenario **Licensing.**

- A new graduate applies, the qualification is verified with the training institution, a licence is issued and the registry is updated.
- Renewal linked to continuing professional development.
- Status changes, such as suspension and reinstatement, with notification to employers.

Personas: Health worker, Training institution registrar, Registration officer (regulatory body), Employer or facility human resources manager

20.2 Verification

User scenario **Verification.**

- An employer or member of the public checks a health worker's status online.
- A clinical system checks automatically that a prescriber is authorized.

- A health worker holds a digital credential that can be verified online or offline.

Personas: Employer or facility human resources manager, Member of the public, Electronic medical record, Health worker

20.3 Emergency response

User scenario **Emergency response.**

- Quickly identify available, qualified workers by skill and location for surge deployment.
- Verify the credentials of emergency medical teams and volunteers.
- Track deployment and return.

Personas: Emergency coordinator, Health worker, Registration officer (regulatory body)

20.4 Mobility

User scenario **Mobility.**

- Verify the credentials of internationally trained workers and issue certificates of good standing.
- Track inflows and outflows to support reporting under the WHO Global Code of Practice on the International Recruitment of Health Personnel.
- Movement within the country between regions, sectors and employers.

Personas: Health worker, Registration officer (regulatory body), Workforce planner

Figures, tools and expected outputs

Suggested figures and tables

- Figure: workflow diagram for each use case.
- Table: people, data and interfaces involved in each use case.

Implementation tools and checklists

- Use case template.
- Scenario test scripts.

Expected outputs for countries

- National use cases adapted for priority scenarios.

Chapter 21. Country Case Studies

Chapter objectives. This chapter will help readers to:

- Present how countries with different governance models have built their registries.
- Draw out practical lessons, including from fragile and low-resource settings.
- Help readers find experience that is relevant to their own context.

Key messages

- Countries have succeeded with different models. Context determines the choice.
- Progress is usually gradual, over several years.
- Governance and data quality are recurring challenges everywhere.
- Fragile settings can start with simple, robust solutions.

21.1 Centralized

- A country with a single national registry across professions (case to be confirmed by the working group).
- How it was established: legal basis, governance and financing.
- Results and challenges.

21.2 Federated

- A country where several councils or sub-national authorities keep their own registers, linked to give a national view (case to be confirmed by the working group).
- How identifiers, the data set and interfaces were harmonized.
- Results and challenges.

21.3 Fragile settings

- A fragile, conflict-affected or very low-resource setting, to be nominated through WHO regional offices.
- A minimum workable approach, options for working offline, and coordination of partners.
- Protecting data and health workers where security is poor.

21.4 Lessons learned

- Common success factors and pitfalls.
- What countries would do differently.
- What the lessons mean for the country pathway.

Figures, tools and expected outputs

Suggested figures and tables

- Summary box for each case study, in a standard format.
- Table: comparison across the case studies.

Implementation tools and checklists

- Case study documentation template.

Expected outputs for countries

- A list of peer countries to learn from.
- The country's own documented experience, to share with others.

Chapter 22. Tools and Templates

Chapter objectives. This chapter will help readers to:

- Provide ready-to-adapt tools for each step of the pathway.
- Explain when and how to use each tool.
- Point to the annexes and online resources where full versions are kept.

Key messages

- Tools save time but must be adapted to the national context.
- Each tool is linked to a step in the country pathway.
- Completed tools should be kept up to date as living documents.
- Tools will be maintained online and, where relevant, in machine-readable form.

22.1 Checklists

- Legal and governance checklist.
- Privacy and security checklist; interoperability checklist.
- Go-live checklist.

22.2 Roadmaps

- Sample phased roadmaps for each starting point: paper, stand-alone digital, and partly connected.
- Milestones, dependencies and decision points.
- Budget and resource planning linked to the roadmap.

22.3 Assessment tools

- Readiness assessment, maturity model and data quality assessment.
- How to carry out each assessment: who takes part, scoring, and use of results.
- Repeating assessments to track progress.

Figures, tools and expected outputs

Suggested figures and tables

- Table: all tools, with purpose, when to use, related chapter and annex.

Implementation tools and checklists

- The full set of tools and templates listed in this chapter and in the annexes.

Expected outputs for countries

- A completed set of adapted tools that together document the country's registry.

Annexes

The annexes hold the full versions of the tools, templates and reference material the chapters point to. Annex 9 (Glossary) is its own chapter.

Annex 1. Minimum Data Set

Proposed core and optional data elements with working definitions.

The elements are drafted as structured data-element blocks in section 11.1.

Annex 2. Data Dictionary

For each data element: definition, data type, permitted values, source, and mapping to FHIR.

Annex 3. Governance Template

Model terms of reference for the steering committee and technical working group, a responsibility matrix, and a sample data-sharing agreement.

Annex 4. User Personas

Full generic persona descriptions for countries to adapt.

The generic personas are drafted as structured persona blocks in section 5.1.

Annex 5. Functional Requirements Catalogue

Numbered functional and non-functional requirements, each with a priority and linked use case.

The functional and non-functional requirements are drafted as structured requirement blocks in Chapter 7.

Annex 6. Interoperability Models

Reference data flows, standard interfaces and example exchanges for centralized, federated and hybrid models.

Annex 7. Readiness Assessment

Questionnaire and scoring guide for each assessment area.

Annex 8. Maturity Model

Levels and criteria for each area, with a self-assessment form.

Annex 10. NHWA Mapping

Table linking registry data elements to NHWA indicators.

Annex 9. Glossary

About this glossary.

This glossary covers health workforce terms. Digital terms are addressed separately.

The working group's task is to agree a definition for each term, staying consistent with existing WHO publications unless there is a reason to update one. The priority terms are *health workforce registry*, *human resources for health information system* and *professional council*. The registry definition must match the final definition in the DPI-H reference architecture.

Each entry below lists the candidate definitions proposed in the TWG glossary workbook (29 September 2026), each with its source. Comment on the entry: say which candidate you prefer, or propose an edited or new definition. Some definitions and sources in the workbook were generated with AI and may need correction.

The machine-readable form is in `glossary/` (SKOS concept schemes). This chapter is generated from it by `scripts/glossary/build_glossary.py`; edit the glossary, not this chapter.

Priority terms for TWG agreement

Health workforce registry.

Also: Health worker registry, Provider registry.

Status: open. 12 candidate definitions, none adopted.

Candidate 1 (WHO/ITU. DPI-H reference architecture, draft v1.0 for public comment. July 2026):

> The Health Workforce Registry is the national authoritative record of identity, credential, and classification data for all health workers recognised as part of the national health workforce — including licensed or certified professionals employed in both public and private sectors and, where nationally defined and registered, community health workers and other cadres operating in informal or semi-formal capacities. It maintains a standardised, uniquely identified record for each health worker, providing the common reference point against which clinical, administrative, financing, and regulatory systems can consistently identify, verify, and classify the people delivering health services.

Candidate 2 (WHO/ITU. DPI-H reference architecture, draft v1.0 for public comment. July 2026):

> The Health Workforce Registry is the national authoritative record of identity, credential, and classification data for all health workers recognised as part of the national health workforce — including licensed or certified professionals employed in both public and private sectors and, where nationally defined and registered, community health workers and other cadres operating in informal or semi-formal capacities. It maintains a standardised, uniquely identified record for each health

worker, providing the common reference point against which clinical, administrative, financing, and regulatory systems can consistently identify, verify, and classify the people delivering health services.

The registry holds two interlocking categories of information: 1. The first is identity: who the health worker is — their demographics, unique identifier, and may cross-reference to foundational identification systems. 2. The second is credentials: what the health worker is qualified and authorised to do — their 240 occupational classification, professional registration, licensing status, and qualifications.

Together, these form the data foundation for a health system that can assure the quality and accountability of its workforce at scale.

The WHO National Health Workforce Accounts (NHWA) handbook (Second Edition, 2023) defines 29 244 indicators that underpin health workforce monitoring, planning, and policy — covering stock, flow, education, migration, and expenditure. The national health workforce registry is a key data source for 12 of 246 these indicators, notably those measuring stock and flow – critical to workforce planning and equitable distribution. The Health Workforce Registry, as defined in the DPI-H reference architecture, provides the foundational data infrastructure from which these indicators can be derived.

Architecture placement: The Health Workforce Registry is a core DPI-H foundational component — the shared identity and credential anchor for health workers across the health ecosystem. It is architecturally distinct from two operational systems: the Human Resources Information System (HRIS), which manages employment transactions, payroll processing, leave management, and workforce scheduling; and the Learning Management System (LMS), which manages training delivery, course enrolment, and assessment. The Health Workforce Registry holds the authoritative master record of who a health worker is and what they are qualified and licensed to do. The HRIS and LMS consume this registry as their reference source; neither replaces it.

Candidate 3 (WHO. HRH information system: minimum data set for health workforce registry. 2015):

"In a national context an electronic health workforce registry can be considered the single authoritative source of health workforce information that can provide an accurate count of all health care personnel that either have worked or are currently working at national or sub-national levels, including in the private sector. An electronic health workforce registry can be designed to authenticate and validate the existence of a health worker and provide essential details about that person." (p. 7) The health workforce registry is a central and essential component of a functional HRIS, as it contains important health worker attributes that can subsequently be linked to other relevant HRIS components." (p8)

"In this document, a minimum data set is proposed to construct an electronic health workforce registry. Using this tool, data elements can be standardized within electronic systems to help facilitate the collection, collation, analysis, reporting and utilization of national and sub-national health workforce data. The tool can also

assist in achieving data interoperability between and within information systems."(p8)

Candidate 4 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> “Regulators generally maintain a registry of regulated practitioners to keep the public informed of those authorized as competent to practice and those who have been suspended or removed from the registry. The level of detail available about practitioners through the regulator’s public website varies between regulators, occupations and countries, from a simple verification of registration to details on qualification, and practice and disciplinary history. Different types of registration exist (for example, student, temporary, provisional, limited practice, supervised practice, private practice, public sector practice, independent practice, specialized practice and nonclinical practice). These determine where a health practitioner can work, their scope of work and level of independence. While in some countries periodic renewals are required to maintain a practitioner’s registration or licence, in others, regulators have limited capacity to maintain a complete registry (p11)
>
> Based on data available from regulators, the registries of regulated health practitioners may include information on the number of students in health occupations, active and inactive health practitioners, their personal details, place of education and employment, and the jurisdiction of registration within a country. (p13)
>
> The inclusion of disciplinary history in public registries can help patients, clients and employers to make informed choices about practitioners. Regulators can use the data to identify and reduce the risk of harm to the public through preventative action or early identification and intervention (p45)
>
> A core function of regulators is to maintain a publicly available registry on regulated health practitioners that is accurate, complete and includes information on practitioners who have been removed from the registry. This enables access to information on practitioners by both the public seeking services and employers recruiting the practitioners. Depending on the occupation and jurisdiction, the information in the public registry can vary. The health practitioner registry accessible to the public is in compliance with legislation and requirements on confidentiality and the handling of personal data and generally includes relevant information on the practitioners such as name, gender, age, registration number, qualification details (such as basic and additional qualifications, conferring institution, year conferred), scope of registration or licensure, disciplinary history and associated sanctions and actions (26, 115, 286–288). Since maintaining an up-to-date public registry has resource implications, the costs and benefits should be assessed for each data point included in the register. Information shared in the public registry is only a subset of the information that is collected by the regulators. Additional information captured by regulators could include practitioner characteristics such as language skills, country of birth, nationality, details on practice location, criminal history records, fitness to practice, and complaints and disciplinary actions. Information on employers may also be captured (26, 289–292). Some of the information held by

> regulators is confidential and the grounds on which this information may be shared and with whom need to be specified by law. Similarly, the regulator for health practitioner education could hold information on the capacity of the institutions as well as annual data on enrolments and graduates (if applicable) in different programmes, including gender, age, country of birth and the nationality of students. Where assessments of practitioner performance standards are regular and linked to periodic registration or licensure renewals, it can help to keep the practitioner registry updated. In such cases, the available data (for example, on education, entry to practice, maintenance of standards, complaints and discipline) should be analysed to detect clinical governance failures that harm patients, identify innovations and generate evidence and data for improving regulatory systems, practice environments, service delivery and patient safety (p46)

Candidate 5 (WHO. Strengthening the collection, analysis and use of health workforce data and information. 2022):

> “Professional registries maintained by regulators: Regulators maintain a registry of health workers who have been authorized to enter into practice. The information on the registry varies between countries and regulators. In some countries the registration or licensure of health workers is time-limited and requires periodic renewals while in others it can be valid for life.”

Candidate 6 (Joachim M et al. Aligning health workforce forecasting and planning with changing care models):

> “Interoperable health professional registers with a clear legal basis underpin workforce intelligence. Effective registers that integrate data from education, licensing and employment systems are essential if countries are to forecast better at the national and regional levels.” (p5)

> "A health professional register is an official, usually mandatory, system that records information on individuals who are authorized to practise a regulated health profession. Registers are typically maintained by professional regulators or public authorities and contain data such as professional qualifications, scope of practice, licensing status and, depending on the system, information on employment, sector, working hours and location." (p. 23)

> "Their primary purpose is to support professional regulation and patient safety, but they can also serve as a critical source of workforce intelligence. Health professional registers represent a critical source of high-quality, routinely updated workforce intelligence. They provide a robust empirical foundation for understanding the size, composition, distribution, and activity patterns of the HCWF and can inform a wide range of policy functions, including workforce forecasting and planning, education and training capacity, regulation, distribution and budgeting." p23.

Candidate 7 (Leitner C et al. Health worker information systems in low resource settings (IntraHealth, 2013)):

> "...a Provider Registry (PR), also called a Healthcare Provider Directory (HPD), which “supports queries against, and management of, healthcare provider information that

may be publicly shared in a directory structure." The scope of the data is national and cross-sectoral but limited to rudimentary provider information. The PR, as part of a Service Oriented Architecture (SOA), is intended to be the canonical source of health worker data for a country. Within the HWIS, it has the potential to serve as the interface mechanism for the various components highlighted in Section 2."

Candidate 8 (OpenHIE Health Worker Registry implementation guide (IntraHealth/CapacityPlus, 2016)):

> Glossary: "Health Worker Registry: a national authority of health worker information according to a defined minimum data set." "Provider Registry: Another synonym for Health Worker Registry." "Health Worker Directory: List of health workers that are managed by individual organizations, which can come together to form a health worker registry."
>
> Narrative definition: "A Health Worker Registry (HWR) serves as the digital, central authority for maintaining the unique virtual identities of health workers within a country. . The Health Worker Registry pulls data from underlying human resource information systems (HRIS), which play the role of Health Worker Directories, and makes these data available as a single virtual database. These Health Worker Directories could include Ministry of Health installations of tools like iHRIS, private systems used by faith-based organizations, and private-sector systems as well as those used by regulatory bodies like nursing councils. These health Worker directories are responsible for specific business processes: maybe one tracks health worker certifications for a professional council or another holds salary information for the ministry of health. The information stored in these directories mirrors their business cases. The Health Worker Registry has no business case other than being a master document of all health worker information that specific directories can pull and share. The Health Worker Registry contains a Minimum Data Set (MDS) of details of all health workers working in both public and private sectors, such as full name, contact information and place(s) of employment.
>
> The Health Worker Registry (HWR) serves as the authority for maintaining the unique identities of health workers within a country (p12)
>
> Governments collect and store multiple and disparate sources of data on health workers. Thus, it is a complex task to pull together and maintain a master list of all health workers in a country. A Health Worker Registry (HWR) seeks to reduce the complexity of this task by: ● creating a Minimum Data Set (MDS) of health workforce information from the various source data systems ● merging the information from a variety of data systems into an authoritative registry of health workers according to a data governance policy ● allowing queries of health worker information by various computer and human users, also called the client systems.
>
> The HWR serves as a publicly available "electronic phone-book" of health workers and health services across all sectors (public, private and faith-based organizations). As the World Health Organization points out, the HWR can be seen as the "single authoritative source" of health workforce information that parts of the HIS can query to obtain details on health workers and automatically populate their databases, rather than having to manually enter those details themselves.

Candidate 9 (IntraHealth Tech Tuesday: health worker registries, MDS and NHWA (2015)):

> "searchable directory of minimal set of health worker information" "no intrinsic management function" "designed to fit in as part of larger health information system." Health Worker Registry Combines Data Sources: pulling together Deployment, License, and Training data from separate source systems.

Candidate 10 (Chibuzor M et al. Implementation of health workforce information systems: eight sub-Saharan countries):

> "HWR or HRH registry is a 'single authoritative source of health workforce information, that can provide an accurate count of all health care personnel that either has worked or are currently working at national or sub-national levels, including in the private sector'" (World Health Organization. Monitoring the building blocks of health systems: a handbook of indicators and their measurement strategies.)

> "The HWR does not function in isolation. It is a part of the Human Resource Information System (HRIS) (also called health workforce information system). HRIS is a system that collects, maintains and analyses data on the health workforce. Other components of HRIS are payroll, training, performance management,migration, benefits and retirement components (Fig. 1). The HWR is, however, a central component of the HRIS and provides data on which all the other components depend.

Candidate 11 (Digital Health Enterprise Planning Approach, training slide deck (undated)):

> "A health worker registry is the central authority for maintaining the unique identities of health providers within the country"

Candidate 12 (WHO Minimum Data Set (MDS) crosswalk (undated)):

> "A health worker registry is a centralized listing of health workers within a country. Information contained within a health worker registry most often comes from different health management information systems (HMIS), such as iHRIS, used, for example, by independent organizations, health clinics run by non-state actors or regulatory bodies. These client systems are often organized around specific workflows revolving around human resources (hiring a new health worker, providing a salary bonus, transferring a health worker to a new location), but the health worker registry may contain only specific information about health workers."

Table.

#	Source	Year	Opening sentence
1	WHO/ITU. DPI-H reference architecture, draft v1.0 for public comment. July 2026	2026	The Health Workforce Registry is the national authoritative record of identity, credential, and

#	Source	Year	Opening sentence
			classification data for all health workers recognised as part of the national health workforce — including licensed or…
2	WHO/ITU. DPI-H reference architecture, draft v1.0 for public comment. July 2026	2026	The Health Workforce Registry is the national authoritative record of identity, credential, and classification data for all health workers recognised as part of the national health workforce — including licensed or…
3	WHO. HRH information system: minimum data set for health workforce registry. 2015	2015	In a national context an electronic health workforce registry can be considered the single authoritative source of health workforce information that can provide an accurate count of all health care personnel that…
4	WHO. Health practitioner regulation: design, reform	2024	Regulators generally maintain a registry of

#	Source	Year	Opening sentence
	and implementation guidance. 2024		regulated practitioners to keep the public informed of those authorized as competent to practice and those who have been suspended or removed from the registry.
5	WHO. Strengthening the collection, analysis and use of health workforce data and information. 2022	2022	Professional registries maintained by regulators: Regulators maintain a registry of health workers who have been authorized to enter into practice.
6	Joachim M et al. Aligning health workforce forecasting and planning with changing care models	—	Interoperable health professional registers with a clear legal basis underpin workforce intelligence.
7	Leitner C et al. Health worker information systems in low resource settings (IntraHealth, 2013)	2013	...a Provider Registry (PR), also called a Healthcare Provider Directory (HPD), which “supports queries against, and management of, healthcare provider information that may be publicly

#	Source	Year	Opening sentence
			shared in a directory structure.”…
8	OpenHIE Health Worker Registry implementation guide (IntraHealth/CapacityPlus, 2016)	2016	Glossary: "Health Worker Registry: a national authority of health worker information according to a defined minimum data set." "Provider Registry: Another synonym for Health Worker Registry." "Health Worker Directory:…
9	IntraHealth Tech Tuesday: health worker registries, MDS and NHWA (2015)	2015	searchable directory of minimal set of health worker information" "no intrinsic management function" "designed to fit in as part of larger health information system." Health Worker Registry Combines Data Sources:…
10	Chibuzor M et al. Implementation of health workforce information systems: eight sub-Saharan countries	—	HWR or HRH registry is a 'single authoritative source of health workforce information, that can provide an accurate count of

#	Source	Year	Opening sentence
			all health care personnel that either has worked or are currently working at national or…
11	Digital Health Enterprise Planning Approach, training slide deck (undated)	—	A health worker registry is the central authority for maintaining the unique identities of health providers within the country"
12	WHO Minimum Data Set (MDS) crosswalk (undated)	—	A health worker registry is a centralized listing of health workers within a country.

Health workforce registry: candidate definitions side by side (opening sentence of each)

Human resources for health information system.

Also: HRH-IS, Health workforce information system, human resources information systems for health, HRIS, HRHIS, Information Systems for Human Resources for Health (ISHRH).

Status: open. 12 candidate definitions, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> An information system that provides, analyses, evaluates and distributes information needed to support decision-making and health personnel management and practices.

Candidate 2 (WHO. Health labour market analysis guidebook. 2021):

> HRH-IS: An information system that provides, analyses, evaluates and distributes information needed to support decision-making and health personnel management and practices. P280

Candidate 3 (WHO. Strengthening the collection, analysis and use of health workforce data and information. 2022):

> A human resources for health information system (HRHIS) is the basic tool that can provide data and information on the health workforce situation and its evolution." p267; An HRHIS enables the monitoring of key information on HRH, such as stock

and demography, education, finances, regulations, etc. It is an integrated element of the broader HIS, not an isolated subsystem."Page: 310

Candidate 4 (Eurohealth. The European health workforce: building a true labour market. 2022):

> “An HRHIS is a subsystem of an HIS, composed of various subsystems such as the payroll information system, education management information system (EMIS), and councils’ and professional associations’ records and information systems.” p261

Candidate 5 (Riley PL et al. Information systems on human resources for health: a global review. 2012):

> "Increasingly, systems for collecting and disseminating information on a country’s health workforce – referred to as human resources information systems (HRIS)... Functional HRIS models involve standardized processes for data capture, management, and use so as to provide accurate, timely, and comprehensive profiles of workforce size, composition, and deployment"

Candidate 6 (Leitner C et al. Health worker information systems in low resource settings (IntraHealth, 2013)):

> "A Human Resource Information System (HRIS)... is one [of] the key components of the HWIS that manages information on current and past employees of a Ministry of Health or other health service delivery organization. An HRIS is concerned with the information related to position/post, job, cadre and other information related to employee management and deployment"

Candidate 7 (Ishijima H et al. HRHIS and TIIS national rollout in Tanzania):

> Describes Tanzania’s "Human Resource for Health Information System (HRHIS)": "HRHIS is a web-based software that enables a health system to collect, validate, analyse and present raw and statistical human resource information for reporting, analysis and decision-making"

Candidate 8 (Were V et al. South-South collaboration on HRIS):

> HRIS collects and manages routine, national level, multi-cadre data on the health workforce including supply (i.e., training, exam, registration, licensure, intent to out-migrate, and continuing professional development) and deployment (i.e., health facility of deployment, date of appointment, workstation in the facility, date of promotion, disciplinary actions, date of exit, and transfers)."

Candidate 9 (Chibuzor M et al. Implementation of health workforce information systems: eight sub-Saharan countries):

> "HRIS is a system that collects, maintains and analyses data on the health workforce. Other components of HRIS are payroll, training, performance management, migration, benefits and retirement components" (p.i28, Fig.1). Data collected covers "the health worker from pre-entry into the workforce to exit at retirement" including

"identification number, name, birth history, employment status, address, education, professional licensure and certification" (p.i28).

Candidate 10 (MEASURE Evaluation, Module 3. 2018):

"A human resources information system (HRIS) provides data to track the status of the workforce in terms of the number, occupation, and geographical distribution of health workers."

Candidate 11 (Waters KP et al. Zimbabwe's HRH information system (ZHRIS)):

"A human resources information system (HRIS) is a system of collection, storage, analysis, and reporting of data to produce meaningful information about the health workforce, including demographics, capacity, training needs, deployment, and migration patterns."

Candidate 12 (Wikipedia: HRHIS):

A human resource for health information system (HRHIS), also known as human resource information system (HRIS) — is a system for collecting, processing, managing and disseminating data and information on human resource for health (HRH). Depending on the level of development of a country's health care system and the organization of its workforce, an HRHIS can be computerized or paper-based, including information on numbers and distribution of health workers and track their career information.[1]

Table.

#	Source	Year	Opening sentence
1	WHO. Health labour market analysis guidebook. 2021	2021	An information system that provides, analyses, evaluates and distributes information needed to support decision-making and health personnel management and practices.
2	WHO. Health labour market analysis guidebook. 2021	2021	HRH-IS: An information system that provides, analyses, evaluates and distributes

#	Source	Year	Opening sentence
			information needed to support decision-making and health personnel management and practices.
3	WHO. Strengthening the collection, analysis and use of health workforce data and information. 2022	2022	A human resources for health information system (HRHIS) is the basic tool that can provide data and information on the health workforce situation and its evolution.” p267;
4	Eurohealth. The European health workforce: building a true labour market. 2022	2022	An HRHIS is a subsystem of an HIS, composed of various subsystems such as the payroll information system, education management information system (EMIS), and councils’ and professional associations’ records and…
5	Riley PL et al. Information systems on human resources for health: a global review. 2012	2012	Increasingly, systems for collecting and disseminating information on a country’s health workforce – referred to as human resources

#	Source	Year	Opening sentence
			information systems (HRIS)...
6	Leitner C et al. Health worker information systems in low resource settings (IntraHealth, 2013)	2013	A Human Resource Information System (HRIS)...
7	Ishijima H et al. HRHIS and TIIS national rollout in Tanzania	—	Describes Tanzania's "Human Resource for Health Information System (HRHIS)": "HRHIS is a web-based software that enables a health system to collect, validate, analyse and present raw and statistical human resource…
8	Were V et al. South-South collaboration on HRIS	—	HRIS collects and manages routine, national level, multi-cadre data on the health workforce including supply (i.e., training, exam, registration, licensure, intent to out-migrate, and continuing professional…
9	Chibuzor M et al. Implementation of health workforce information	—	HRIS is a system that collects, maintains and analyses data on the health

#	Source	Year	Opening sentence
	systems: eight sub-Saharan countries		workforce.
10	MEASURE Evaluation, Module 3. 2018	2018	A human resources information system (HRIS) provides data to track the status of the workforce in terms of the number, occupation, and geographical distribution of health workers."
11	Waters KP et al. Zimbabwe's HRH information system (ZHRIS)	—	A human resources information system (HRIS) is a system of collection, storage, analysis, and reporting of data to produce meaningful information about the health workforce, including demographics, capacity, training…
12	Wikipedia: HRHIS	—	A human resource for health information system (HRHIS), also known as human resource information system (HRIS) — is a system for collecting,

#	Source	Year	Opening sentence
			processing, managing and disseminating data and information on human resource…

Human resources for health information system: candidate definitions side by side (opening sentence of each)

Professional council.

Also: professional regulatory/licensing authority/body, medical council/board, nursing council/board.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> Organization with the legal authority to regulate one or more health occupations with a view to protecting the public. Its mandate can cover the regulation of entry into the occupation (education requirements), the monitoring and promotion of the quality of practice, and administration of discipline... At national level (and at subnational level where legislative power exists), examples of mandatory rules are professional laws that define the powers and duties of professional councils, such as registration, certification of specialties and discipline. They also define scopes of practice that, in turn, shape the division of labour in the HLM.

Candidate 2 (OpenHIE Health Worker Registry implementation guide (IntraHealth/CapacityPlus, 2016)):

> A national, legislatively mandated entity that certifies one or more cadres of health workers as meeting predetermined professional requirements and standards of practice.

Country examples are listed in the section *Professional councils: country examples*.

Table.

#	Source	Year	Opening sentence
1	WHO. Health labour market analysis guidebook. 2021	2021	Organization with the legal authority to regulate one or more health occupations with a view to

#	Source	Year	Opening sentence
			protecting the public.
2	OpenHIE Health Worker Registry implementation guide (IntraHealth/CapacityPlus, 2016)	2016	A national, legislatively mandated entity that certifies one or more cadres of health workers as meeting predetermined professional requirements and standards of practice.

Professional council: candidate definitions side by side (opening sentence of each)

Health workforce terms

Accreditation.

Term: Accreditation.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> In a health workforce context, the evaluation of educational institutions and programmes of study against pre-defined standards required for the delivery of education. The outcome of the process is certification of the suitability of education programmes, and the capability of educational institutions to deliver initial and/or continuing education.

Active health worker.

Term: Active health worker.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> One who provides services to patients and communities (practising health worker) or whose medical education is a prerequisite for the execution of the job (e.g. education, research, public administration) even if the health worker is not directly providing services (professionally active health worker). If data are not available for practising or professionally active health workers, data with the closest definition can be used, such as “health worker licensed to practice”. Categories of level are based on the

definitions of OECD/Eurostat/WHO-Europe Joint questionnaire on nonmonetary health care statistics.

Activity level.

Term: Activity level.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

• Practising: This includes all health and care personnel directly providing services to or for patients and communities. • Professionally active: This includes practising health and care personnel as well as other personnel who are directly not providing services to patients but for whom their medical or paramedical education is a prerequisite for the execution of the job (e.g. education, research, public administration). • Licensed to practice: This includes all health and care personnel who are registered and entitled to practice.

Attrition.

Term: Attrition.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

Defined broadly as exits from the workforce, which can be due to emigration, voluntary exits (for example, to other sectors of employment), illness, death or retirement

Certification.

Term: Certification.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

The recognition that an individual has met certain qualification requirements.

Community health workers.

Term: Community health workers.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

Health workers who provide health education, referral and follow-up, case management, basic preventive health care and home visiting services to specific communities. They provide support and assistance to individuals and families in navigating the health and social services system.

Competence.

Term: Competence.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> The state of a person's proficiency to perform the required work activities to the defined standard. This includes having the requisite competencies in each context. Competence is multidimensional, dynamic and changes with time, experience and setting

Competencies.

Term: Competencies.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The abilities of a person to integrate knowledge, skills, and attitudes in their performance of tasks in a given context. Competencies are durable, trainable and, through the expression of behaviours, measurable.

Continuing professional development (CPD).

Also: CPD.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> Training that is beyond clinical update and includes wide-ranging competences like research and scientific writing; multidisciplinary context of patient care; professionalism and ethical practice; communication, leadership, management and behavioural skills; team building; information technology; auditing; and appropriate attitudinal change to ensure improved patient service, research outcomes, and attainment of the highest degree of satisfaction by stakeholders.The form of continuing professional development (CPD) may include: courses and lectures; training days; peer review; clinical audit; reading journals; attending conferences; e-learning activity. CPD may be included in national standards of conduct, performance and ethics that govern health workers.

Candidate 2 (WHO. Health labour market analysis guidebook. 2021):

> All educational experiences, activities and resources engaged by a health professional after completion of professional training. The form of continuing professional development may include courses and lectures, training days, peer review, clinical audit, reading journals, attending conferences, or e-learning activity. National systems for continuing professional development may be voluntary or mandatory.

Contract type.

Term: Contract type.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> The contract type is a key employment characteristic of all workers, including health workers. Personnel may be categorized as full-time workers or part-time workers, depending on their specific contract types. Part-time workers refers to those employed persons whose normal hours of work (calculated weekly or on average over a given period of employment) are less than those of comparable full-time workers.

Deployment.

Term: Deployment.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The process of allocating personnel among types and levels of services and among regions and subregions of a country.

Distribution.

Term: Distribution.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The result of deployment. For example, balanced distribution refers to the appropriate allocation of health personnel, geographically and among levels of care and types of services, to ensure equitable provision of quality health services to all.

Dual practice.

Term: Dual practice.

Status: open. 3 candidate definitions, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> Phenomenon in which a health practitioner is simultaneously employed or holds dual or multiple appointments (in both the public and private sectors) in regulated as well as unregulated ways and in an authorized or informal manner.

Candidate 2 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> Multiple health-related practices by a health worker in the same or different sites. Dual practice in this sense may be public-on-public, public-on-private, or private-on-private.

Candidate 3 (WHO. Health labour market analysis guidebook. 2021):

> There are several forms of dual practice. Health professionals can work in a public service provision role and another role: (a) outside: in a completely separate private environment; (b) beside: in a private ward or clinic physically associated with a public facility but run as a separate business; (c) within: where private services are offered inside a public facility but outside public service operating hours or space; or (d) integrated: where additional fees are charged for services offered alongside standard public ones, often informally, on the understanding of a faster – or higher-quality – service. Academics and policy-makers typically restrict the term dual practice to category (a), but it is clear that categories (b), (c) and (d) present alternative scenarios for health professionals to combine public and private practice, and supplement public sector salaries; policy-making in this domain should navigate the trade-offs between the objective of retaining personnel while ensuring their commitment to public sector objectives (15)

Education (for health workers).

Term: Education (for health workers).

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The process of developing knowledge, skills, attitudes and competencies related to the delivery of health services. Specialization is the process of developing advanced knowledge, skills, attitudes and competencies related to the delivery of specific health services.

Educational institution.

Term: Educational institution.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> An established institution that provides education as its main purpose, such as a school, college, university or training centre. Such institutions are normally accredited or sanctioned by the relevant national education authorities or equivalent to award qualifications. Educational institutions may also be operated by private organizations, such as religious bodies, special interest groups or private educational and training enterprises, both for-profit and non-profit.

Education and training institution (Health workforce).

Term: Education and training institution (Health workforce).

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> An established institution that provides education as its main purpose, such as a school, college, university or training centre. Such institutions are normally accredited or sanctioned by the relevant national education authorities or equivalent to award qualifications. Educational institutions may also be operated by private organizations, such as religious bodies, special interest groups or private educational and training enterprises, both for-profit and not-for-profit. Types of health workforce education and training institutions are: • Public: Public educational institutions provide core educational services such as teaching activities and ancillary services. They include schools, colleges, universities, and training centres. They are controlled and managed directly by a public education authority or governing body (council, committee, etc.), the majority of whose members are appointed by a public authority. • Private: Private educational institutions provide core educational products such as teaching activities and ancillary services. They include schools, colleges, universities, and training centres, that are controlled and directly managed either by a private organization such as a church, trade union, or business enterprise, or by a governing board whose members have mostly not been selected by a public authority. Whether or not an institution is private is therefore a matter of management, not funding. A school, for example, could in theory be entirely publicly funded but still be considered private because it is not managed by the government. In practice, for international comparability, any educational institution not managed by a government institution is classified as private.

Employment status: full-time, part-time, temporary, permanent.

Term: Employment status: full-time, part-time, temporary, permanent.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> Full-time (whole-time) is employment for or working for the amount of time considered customary or standard. Part-time is employment for or working for less than the amount of time considered customary or standard. Permanent is employment contracted for an indeterminate period. Fixed term is employment contracted for a fixed period of time. Temporary refers to short-term contracts or "casual" work, either for a definite period or for a specific activity. Self-employment is when remuneration is directly dependent upon the profits derived from the goods and services produced by the individual.

Health labour market.

Term: Health labour market.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO AFRO. Africa Health Workforce Investment Charter):

> The structure that allows the services of health workers to be sought (demanded) and offered (supplied). The health labour market can be characterized according to geographical area (local, national or international); occupation (by occupation title or category, specialized or unspecialized); and sector (private or public, formal or informal). The dynamic between the number and the kind of jobs offered on the market and the number of health workers is central in determining the configuration of the health labour market

Health practitioner.

Term: Health practitioner.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> For the purpose of this document, the term “health practitioner” encompasses all health professionals, health associate professionals, including community health workers, health care assistants, and personal care workers in health services (allopathy as well as traditional, complementary and integrative medicine) as defined in the International Standard Classification of Occupations (ISCO-08), and new or health practitioners that are yet to fall under an official classification but who are directly involved in patient diagnostics or care

Health professional.

Term: Health professional.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> Health personnel who apply knowledge, such as those relating to medicine, nursing, midwifery, dentistry, and allied health and health promotion; they usually require a university undergraduate or postgraduate degree or the equivalent

Health Professions Education Accreditation Agency.

Term: Health Professions Education Accreditation Agency.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Framework for the Global Digital Health Certification Network (GDHCN). 2026):

> An organization or organizational unit that is responsible for establishing and/or implementing procedures for accreditation of health professions education institutions4.

Health Services Licensing Agency.

Term: Health Services Licensing Agency.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Framework for the Global Digital Health Certification Network (GDHCN). 2026):

> An organization or organizational unit that is responsible for establishing and/or implementing procedures for licensure, and the licensing of organizations or individuals to provide clinical, health system or public health services within their jurisdiction. Licensing comprises processes through which duly authorized governmental authorities, such as recognized professional organizations, grant permission to an individual or healthcare organization to operate or engage in a medical occupation or profession.

Health workers.

Also: health workforce, human resources for health (HRH), HRH.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO WPRO. Regional framework to shape a health workforce for the future of the Western Pacific. 2024):

> All persons engaged in actions whose primary intent is to enhance health (3,5).

Health workforce planning.

Term: Health workforce planning.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> Strategies that address the adequacy of the supply and distribution of the health workforce according to policy objectives and the consequential demand for health labour. (63)

Candidate 2 (WHO. Health labour market analysis guidebook. 2021):

> The process of estimating potential requirements for human resources for health and of designing ways of fulfilling those requirements, including strategies that address the adequacy of the supply and distribution of the health workforce according to policy objectives and the consequential demand for health labour (8)

Health workforce (HWF).

Also: HWF.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO AFRO. Africa Health Workforce Investment Charter):

> All people engaged in actions whose primary intent is to enhance health. They may be paid staff or volunteers working full-time or part-time in the public and private sectors. They may be delivering health services, managing the services offered by the system, or addressing social determinants of health. That means that the health workforce includes: all personnel trained in health occupations delivering clinical work in health facilities (such as medical doctors, nurses, or dentists); all non health professionals employed in the health sector, public and private, regardless of their occupation (such as managers, ambulance drivers or teachers of health education); and all those whose work supports the delivery of health services, even if they are employed by other sectors or industries (such as cleaning, catering, security or agency staff working in the health sector)

Health workforce observatory.

Also: HRH observatory (national).

Status: open. 4 candidate definitions, none adopted.

Candidate 1 (WHO AFRO. Regional framework for Workforce 2030. 2020):

> A Health Workforce Observatory is a platform or mechanism for generating, analyzing, sharing, and using human resources for health (HRH) data and evidence to support policy-making, planning, monitoring, and research. The African Regional Framework describes national health workforce observatories as platforms for generating and sharing HRH data and best practices and notes that workforce observatories support evidence-based policies

Candidate 2 (WHO. Strengthening the collection, analysis and use of health workforce data and information. 2022):

> Where they exist, HRH observatories are an excellent governance platform for the coordination of NHWA implementation. These observatories can be expanded and adapted to align progressively with NHWA implementation principles and requirements." Page: 259 "In the WHO African, Americas and South-East Asia regions, several countries mutualize forces through HRH observatories to collect, analyse and disseminate health workforce data. These initiatives resulted from understanding that no single organization or body can compile all the relevant health workforce indicators without overburdening countries and duplicating efforts already conducted by others." Page: 263

Candidate 3 (WHO. Human resources for health observatories. 2012):

> There is no consensual definition of what an observatory is. There are various ways and means to achieve the same goals and objectives, and no blue-print or "best practice" can be prescribed (Hemmings, Wilkinson 2003). In the social sector, these organizations go beyond observing and analysing; they also pro-actively intervene on the reality, which they monitor. The simple activity of data and information-gathering and dissemination has effects on the reality of the sector and on the actors' behaviours. HRH observatories are no exception, and in this sense they are rather more like independent policy think-tanks. It is therefore important to note first that

there are various “types” of HRH observatory, which take different forms depending on when they were established, by whom, where they are located organizationally, what functions and activities they perform, which resources and funding sources they have, and on the actors involved in their development and management. Table 1 presents that variety of types of observatories according to a number of significant dimensions.

Candidate 4 (OpenHIE Health Worker Registry implementation guide (IntraHealth/CapacityPlus, 2016)):

> A cooperative organization designed to promote, develop and sustain a knowledge base for human resources for health in a region or country. It provides evidence for policy decisions to strengthen health systems and improve health service delivery."

Human resources management.

Term: Human resources management.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The process of creating an adequate organizational environment and management of health workers, including hiring, defining incentive systems, performance monitoring, supervision, and career development

In-service training.

Term: In-service training.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> Training received while one is fully employed in the health sector. The aim is to equip health workers or the trainers of health workers with the skills to deliver specific interventions. (69

Candidate 2 (WHO. Health labour market analysis guidebook. 2021):

> Maintenance and adaptation of the competencies of existing personnel within the context of their current position; training received while one is employed in the health sector (8)

International Standard Classification of Occupations (ISCO).

Also: ISCO.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

An international classification for organizing jobs into a clearly defined set of groups according to the tasks and duties undertaken in the job. It is maintained by the International Labour Organization.

Labour force survey.

Term: Labour force survey.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

A household-based sample survey focused on the labour force status of the working age population and related statistics. Such surveys often allow disaggregation of the labour force by personal characteristics such as sex, age, educational attainment, and, in some cases, by migrant status and ethnicity, as well as providing information about the jobs held by employed persons (such as occupation and type of contract).

Licensing.

Term: Licensing.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO WPRO. Regional framework to shape a health workforce for the future of the Western Pacific. 2024):

A process by which a governmental authority grants permission to an individual practitioner or health-care organization to operate or to engage in an occupation or profession (4).

Candidate 2 (WHO. Health labour market analysis guidebook. 2021):

Licensure is a process by which a governmental authority grants permission to an individual practitioner or health care organization to operate or to engage in an occupation or profession. Licensure regulations are generally established to ensure that an organization or individual meets minimum standards to protect public health and safety. Licensure to individuals is usually granted after some form of examination or proof of education and may be renewed periodically through payment of a fee and/or proof of continuing education or professional competence (11).

Migration.

Term: Migration.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> Process of movement of people from one country, region or place to settle in another. Moving out is emigration; moving in is immigration. The expressions of outflow and inflow are also used

Mobility flow.

Term: Mobility flow.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The movement of personnel between positions, organizations and regions, or from the public to the private sector within a country, or out of the health sector

National Health Workforce Accounts (NHWA).

Also: NHWA.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> The NHWA can be defined as – "a system by which countries progressively improve the availability, quality, and use of data on health workforce through monitoring of a set of indicators to support achievement of Universal Health Coverage, Sustainable Development Goals and other health objectives".

Occupation.

Term: Occupation.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> A set of jobs with main tasks and duties that are characterized by a high degree of similarity

Candidate 2 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> The concept of occupation is defined as a "set of jobs whose main tasks and duties are characterized by a high degree of similarity". A job is defined as "a set of tasks and duties performed, or meant to be performed, by one person, including for an employer or in self-employment". The National Health Workforce Accounts covers health and health-related occupations grouped according to the International Standard Classification of Occupations 2008.

Place of training.

Term: Place of training.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> This is defined as the place of first qualification for the health worker. Personnel may be categorized as either domestic-trained, foreign-trained, or their place of training may be unknown. The foreign trained health personnel may also be disaggregated into “foreign-trained national-born” (people born in a country who went to study in another country but have come back afterwards to practice in their home country) and “foreign-trained foreign-born”. (101)

Pre-service education.

Term: Pre-service education.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. National health workforce accounts: a handbook, 2nd ed. 2023):

> All mandatory educational activities occurring before graduates enter into service, i.e. excluding all education and training activities done in-service and continuing professional development

Qualification.

Term: Qualification.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> Official confirmation, usually in the form of a document certifying the successful completion of an educational programme or a stage of a programme. Qualifications can be obtained through: (1) successful completion of a full programme; (2) successful completion of a stage of a programme (intermediate qualifications); or (3) validation of acquired knowledge, skills and competencies that is independent of participation in such programmes.

Recruitment.

Term: Recruitment.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> Process of searching for personnel to enter a particular job or position.

Registration.

Term: Registration.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> Official recording of the names of persons who have certain qualifications to practise a profession or occupation

Candidate 2 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> "Registration and licensure are used synonymously in many countries, but in others they mean different things." "Under a statutory registration scheme, the provider's name is placed on a list of people who are considered qualified to provide a service and they are then entitled to use a reserved professional title.

Regulation.

Term: Regulation.

Status: open. 4 candidate definitions, none adopted.

Candidate 1 (WHO. Health practitioner regulation: design, reform and implementation guidance. 2024):

> All laws or rules that govern an individual's entry to health practitioner education programmes, entry to practice, registration, licensure, scope of practice, maintenance of practice standards, disciplinary actions for deviation from regulatory standards, and associated linkages with the health system – adapted from (9).

Candidate 2 (WHO guideline on health workforce development, attraction, recruitment and retention in rural and remote areas. 2021):

> Can be defined broadly to encompass any government control exercised through legislative, administrative, legal or policy tools

Candidate 3 (WHO WPRO. Regional framework to shape a health workforce for the future of the Western Pacific. 2024):

> Any government control exercised through legislative, administrative, legal or policy tools (3).

Candidate 4 (WHO. Health labour market analysis guidebook. 2021):

> The definition of rules that will govern the structure and functioning of education institutions and of service provider organizations and the professional activities of individual health workers.

Retention (of health workers).

Term: Retention (of health workers).

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> Maintenance of health personnel in a specific position or organization, the health sector, region, or country

Scope of practice.

Term: Scope of practice.

Status: open. 2 candidate definitions, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The set of tasks that members of a specific occupation are authorized to perform

Candidate 2 (WHO guideline on health workforce development, attraction, recruitment and retention in rural and remote areas. 2021):

> the health care services an individual health care worker is authorized to perform by virtue of professional licence, registration or certification...” (p. 35)

Stewardship.

Term: Stewardship.

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO WPRO. Regional framework to shape a health workforce for the future of the Western Pacific. 2024):

> A responsibility for the effective planning and management of health resources to safeguard equity, population health and well-being (2)

Stock (of health workers).

Term: Stock (of health workers).

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

> The total number of health workers potentially available in a country, including those participating in the health labour market, plus those who are qualified to do so but do not participate for some reason, such as early retirement.

Supply (of health workers).

Term: Supply (of health workers).

Status: open. 1 candidate definition, none adopted.

Candidate 1 (WHO. Health labour market analysis guidebook. 2021):

The number of health workers active in the health labour market, either in employment or not employed but willing to work.

Professional councils: country examples

Table.

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
AFR	Ghana	Medical and Dental Council (MDC) Ghana	Medicine and dentistry	A statutory agency established under Part II of the Health Professions Regulatory Bodies Act, 2013 (Act 857) to regulate the training and practice of medicine and dentistry.	Secures the highest standards in training and practice in the public interest; determines adequacy of service facilities; prescribes codes of professional conduct; registers local and foreign-trained practitioners.	Health Professions Regulatory Bodies Act, 2013 (Act 857)	**not verified** (sources could not be opened). Suspected: Search snippets are consistent with the row: Part II of Act 857, object in s.27. The MDC also oversees physician assistant training (snippet).

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							Reference gives no URL; suggest adding the GhaLII text of Act 857. Leads: 1 2
AFR	Ghana	Nursing and Midwifery Council of Ghana (N&MC)	Nursing and midwifery	A statutory body established under Part III of the Health Professions Regulatory Bodies Act, 2013 (Act 857), which replaced the earlier Nurses and Midwifery Council legislation (NRCD	Secures, in the public interest, the highest standards of training and practice of nursing and midwifery; indexes, registers, verifies and licenses nurses and midwives; sets and examines	Health Professions Regulatory Bodies Act, 2013 (Act 857); https://www.nmc.gov.gh/web/who-we-are	**not verified** (sources could not be opened). Suspected: The snippet supports Part III and the NRCD 117 predecessor. It also suggests two errors to check once a fetch works. First, the

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				117 of 1972 and LI 683), operating alongside the Medical and Dental Council under the same umbrella Act.	against professional standards.		former name was 'Nurses and Midwives Council', not 'Nurses and Midwifery Council'. Second, 'LI 683 is also being reviewed', so saying Act 857 replaced LI 683 may be inaccurate. The workbook URL www.nmc.gov.gh is the www host, but search indexes nmc.gov.gh withou

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							t www, so the link may not resolve as given. The repeal wording in Act 857 itself is not confirmed. Leads: 1
AFR	Kenya	Kenya Medical Practitioners and Dentists Council (KMPDC)	Medicine and dentistry	A state corporation established under the Medical Practitioners and Dentists Act, Chapter 253 of the Laws of Kenya. (Nursing/midwifery is regulat	Regulates training, registration, licensing, inspection and professional practice of medical practitioners and dentists to ensure quality and ethical	Medical Practitioners and Dentists Act, Cap. 253; https://kmpdc.go.ke/resources/Medical%20Practitioners%20and%20Dentists%20Act%20(2	**not verified** (sources could not be opened). Suspected: No fetch possible. Snippets are consistent with Cap 253, the 2019 rename from Board

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				ed separately — see the Nursing Council of Kenya row added below.)	healthcare provision; licenses and inspects health facilities.	019%20Revision).pdf; https://en.wikipedia.org/wiki/Medical_Practitioners_and_Dentists_Board	to Council, and facility licensing. The mandate also covers community oral health officers (snippet). The Wikipedia URL should be dropped: its title is the pre-2019 'Medical Practitioners and Dentists Board', and it is not authoritative. Replace it with the

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							Kenya Law URL. Leads: 1 2 3
AFR	Kenya	Nursing Council of Kenya (NCK)	Nursing and midwifery	A statutory body established by the Nurses Act, Chapter 257 of the Laws of Kenya, functioning as the sole professional regulator for nursing and midwifery cadres in Kenya, distinct from the Kenya Medical Practitioners and Dentist	Registers and licenses nurses and midwives trained domestically or internationally; conducts national licensing examinations; maintains official registers; approves training institutions and curricula; conduc	Nurses Act, Cap. 257; https://ncckenya.com/beta/about/	**not verified** (sources could not be opened). Suspected: No fetch possible. Likely error, to confirm on fetch: Cap. 257 is now titled the 'Nurses and Midwives Act' (Act 3 of 1983), not the 'Nurses Act'. Search found no 2025

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				s Council.	ts disciplinary proceedings and advises government on nursing matters.		replacement Act. The workbook URL nckenya.com/beta/about/ was not reachable here. Leads: 1 2
AFR	Nigeria	Medical and Dental Council of Nigeria (MDCN)	Medicine and dentistry	A statutory federal regulatory body established under the Medical and Dental Practitioners Act (originally Decree No. 23 of 1988, now Cap. M8, Laws of	Determines the standards of knowledge and skill required for admission to the medical/dental professions; establishes and maintains the register of practitioners entitled to	Medical and Dental Practitioners Act, Cap. M8 (2004); https://www.mdcn.gov.ng/page/about-us/mdcn-mandate-mission-vission	**not verified** (sources could not be opened). Suspected: No fetch possible. Snippets suggest a mandate covering medicine, dentistry and alterna

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				the Federation of Nigeria 2004). (Nursing/midwifery is regulated separately — see the NMCN row added below.)	practise, publishing lists of qualified practitioners; sets codes of professional conduct; regulates related fields such as laboratory and alternative medicine practice.		tive medicine, so the 'professions' field (medicine and dentistry) may be incomplete. The workbook's mention of 'laboratory' practice is not supported by any snippet seen. The official page uses mdcn.gov.ng; the www host in the workbook URL did not resolve

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							here. Decree 23 of 1988 is supported only by a secondary snippet. Leads: 1
AFR	Nigeria	Nursing and Midwifery Council of Nigeria (NMCN)	Nursing and midwifery	A category B parastatal of the Federal Ministry of Health, established under Decree No. 89 of 1979, now the Nursing and Midwifery (Registration, etc.) Act, Cap. N143, Laws of the	Maintains official registers of qualified nursing and midwifery practitioners; conducts registration interviews for foreign-trained professionals; issues three-year professional practic	Nursing and Midwifery (Registration, etc.) Act, Cap. N143; https://nmcn.gov.ng/who-we-are-nmcn/	**not verified** (sources could not be opened). Suspected: No fetch possible. The snippets support Cap N143, still in force. Neither 'Decree No. 89 of 1979' nor 'category B parastatal' appear

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Federation of Nigeria, operating separately from the Medical and Dental Council of Nigeria.	e licences; awards professional certificates on completion of training; can revoke or restore credentials.		ed in any snippet. The 3-year licence comes only from a community wiki. Check whether the 2021 amendment bill received assent. Leads: 1 2
AFR	South Africa	Health Professions Council of South Africa (HPCSA)	Medicine and 11 other health professions (nursing and pharmacy are regulated separately by their own statutory	A statutory health regulatory body and a juristic person established under Section 2 of the Health Professions Act	Controls education, training, and registration for health professions; ensures practitioners maintain professional	Health Professions Act, 1974 (Act No. 56 of 1974)	**not verified** (sources could not be opened). Suspected: No fetch possible. The statute (s.2, Act 56 of 1974) is consist

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
			councils)	(Act No. 56 of 1974).	ethics; enforces compulsory Continuing Professional Development (CPD).		ent with the snippets. Likely error in 'professions': the HPCSA has 12 professional boards, one of them the Medical and Dental board, and these cover many more than '11 other health professions'. Board count is not the same as profession count, so rephrase, for

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							example 'medicine, dentistry and other health professions under 12 professional boards'. No URL is given; add the acts.co.za section 2 text or the HPCSA site. Leads: 1 2
AFR	South Africa	South African Nursing Council (SANC)	Nursing	An autonomous statutory body established under the Nursing Act, 2005 (Act No. 33	Registers nurses and maintains a public register of qualified practitioners; determi	Nursing Act, 2005 (Act No. 33 of 2005); https://www.sanc.co.za/about-sanc/	**not verified** (sources could not be opened). Suspected: No fetch possible. The mandat

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				of 2005), operating separately from the Health Professions Council of South Africa (HPCSA), which does not regulate nursing.	nes the scope of practice and conditions of nursing practice; accredits nursing education institutions and programmes; investigates complaints and imposes disciplinary sanctions, publishing sanctions within 30 days.		e, including the 30-day Gazette publication, is consistent with s.4 of the snippet. Wording to check: SANC predates the 2005 Act (Nursing Act 1944) and continues under s.2 of it, so 'established under' should read 'continues to exist under'. The 'professions'

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							field likely omits midwifery, which the Nursing Act categories include; not confirmed. The sanc.co.za URL was not reachable here. Leads: 1 2
AMR	Canada (Ontario)	College of Nurses of Ontario (CNO)	Nursing (Registered Nurses and Registered Practical Nurses)	Ontario's statutory self-regulatory body for nursing, established under the Nursing Act, 1991, operati	Registers Registered Nurses (RNs) and Registered Practical Nurses (RPNs) across defined registration classes;	Nursing Act, 1991; Regulated Health Professions Act, 1991; https://neltoolkit.rnao.ca/sites/default/files/Legislation%20an	**not verified** (sources could not be opened). Suspected: Controlled acts are authorized under Nursing Act

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				ng together with the umbrella Regulated Health Professions Act, 1991 (RHPA); came into effect in 1993.	defines the profession's scope of practice and protects nursing titles; authorizes controlled acts; operates a Quality Assurance Program and investigates professional misconduct.	d%20Regulation%20An%20Introduction%20to%20the%20Nursing%20Act,%201991_Revised%20June%202014.pdf	s.4 (renumbered s.4.1 per a 2023 amendment note) - consistent with mandate. RNAO PDF could not be resolved. Leads: 1 2 3
AMR	Canada (Ontario)	College of Physicians and Surgeons of Ontario (CPSO)	Medicine	A statutory self-regulatory body operating under the Regulated Health	Regulates the practice of medicine in the public interest; issues certificates of registra	Regulated Health Professions Act, 1991	**not verified** (sources could not be opened). Suspected: none Leads: 1

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Professions Act, 1991 (RHPA) and the Medicine Act, 1991.	tion (licences); monitors ongoing competence; investigates public complaints.		
AMR	United States	Federation of State Medical Boards (FSMB)	Medicine (coordinating body; actual licensure is issued by individual state medical boards, not FSMB itself)	A non-profit national coordinating organization for the ~70 state and territorial medical boards that hold the actual statutory licensing authority for physicians in	Supports state medical boards with shared services, data, and technology (e.g. co-develops the USMLE Step 3 exam with the National Board of Medical Examiners, credent	FSMB Bylaws & individual state Medical Practice Acts; https://www.fsmb.org/about-fsmb/; https://en.wikipedia.org/wiki/Federation_of_State_Medical_Boards	**not verified** (sources could not be opened). Suspected: fsmb.org reference URL unresolved from this session. Leads: 1 2 3

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				the United States (there is no single federal medical licensing body, mirroring NCSBN's role for nursing).	ials verification, uniform applications); promotes regulatory best practice and information-sharing among state boards; does not itself register or license physicians.		
AMR	United States	National Council of State Boards of Nursing (NCSBN)	Nursing (national coordinating body; licensure itself is issued by individual	An independent, non-profit organization through which boards of nursing act and	Develops the NCLEX-RN and NCLEX-PN licensing examinations; maintains the Nursys databas	NCSBN Organizational Bylaws & State Nurse Practice Acts	**not verified** (sources could not be opened). Suspected: Reference gives no URL;

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
			state boards of nursing)	counsel together on matters of common interest.	e for nurse licensure verification; assists state boards in coordinating policy.		suggest https://www.ncsbn.org. Leads: 1 2
AMR	Brazil	Federal Council of Medicine (Conselho Federal de Medicina - CFM)	Medicine	A federal autarchy (autonomous public body) created by Decree-Law No. 7.955/1945 and regulated by Law No. 3.268/1957.	Serves as the supreme regulatory authority for medical practice; issues the Medical Code of Ethics; supervises regional councils (CRMs); registers and licenses physici	Federal Law No. 3.268/1957	**not verified** (sources could not be opened). Suspected: Statute facts consistent across Câmara, Senado and CFM copies per search results. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
					ans nationally.		
AMR	Brazil	Federal Council of Nursing (Conselho Federal de Enfermagem - COFEN)	Nursing	An autonomous federal entity created together with the state-level Regional Nursing Councils (CORENs) by Law No. 5.905 of 12 July 1973, forming a two-tier disciplinary and registration structure paralleling the medical CFM/CRM	Serves as the national disciplinary body for the nursing profession; issues professional identification credentials with legal effect nationwide; sets the Code of Nursing Deontology; regional COREN councils handle most registration and	Law No. 5.905/1973; https://www.cofen.gov.br/lei-n-590573-de-12-de-julho-de-1973/; https://www.planalto.gov.br/ccivil_03/leis/l5905.htm	**not verified** (sources could not be opened). Suspected: none Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				system already in this table.	routine discipline, with COFEN retaining authority over the most serious sanctions.		
AMR	Chile	Superintendence of Health (Superintendencia de Salud)	All regulated health professions (unified national registry, not profession-specific)	A decentralized government regulatory body operating under the Ministry of Health, governed by DFL No. 1 of 2005; functions as a unified, cross-profession registrar rather	Manages the National Registry of Individual Health Providers (Registro Nacional de Prestadores Individuales de Salud) covering all regulated health professions;	Decreto Fuerza de Ley (DFL) N° 1 de 2005	**not verified** (sources could not be opened). Suspected: none Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				than a profession-specific council (Chile's medical and nursing colegios are voluntary professional associations, not statutory licensing bodies).	verifies professional qualifications; oversees public safety standards.		
EMR	Pakistan	Pakistan Medical and Dental Council (PMDC)	Medicine and dentistry	The statutory regulatory authority maintaining the official register of medical and dental practitioners in	Sets minimum qualification and education standards for medicine and dentistry jointly with the Higher Educati	Pakistan Medical and Dental Council Act, 2022; https://en.wikipedia.org/wiki/Pakistan_Medical_and_Dental_Council; https:/	**not verified** (sources could not be opened). Suspected: evidence is from WebSearch excerpts only. Gazette lists the

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Pakistan, most recently re-established under the Pakistan Medical and Dental Council Act, 2022, after a 2019-2023 period during which its functions were held by the short-lived Pakistan Medical Commission. (Nursing is regulated separately — see the	on Commission; requires registration of all practitioners and students before legal practice; administers the national licensing/registration examination and register.	/pmdc.pk/Documents/law/PMDC%20Act.pdf	2022 Act as Act No. IV of 2023. HEC co-role not found. Leads: 1 2 3

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Pakistan Nursing Council row added below.)			
EMR	Pakistan	Pakistan Nursing Council (PNC) / Pakistan Nursing and Midwifery Council (PNMC)	Nursing, midwifery, lady health visitors (LHVs) and nursing auxiliaries	Pakistan's statutory nursing regulator, founded 1948 with statutory authority under the Pakistan Nursing Council Act (1952, revised 1973); more recently operating as the Pakistan Nursin	Licenses nurses, midwives, lady health visitors and nursing auxiliaries to practise throughout Pakistan; accredited by, and coordinates nursing education quality assurance with, the Higher Education Commission.	Pakistan Nursing Council Act (1952, rev. 1973); https://en.wikipedia.org/wiki/Pakistan_Nursing_Council; https://pnmc.gov.pk/	**not verified** (sources could not be opened). Suspected: 1952 Act not located in primary form. Pending 2025/2026 consolidating PNMC legislation should be re-checked. Leads: 1 2 3 4

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				g and Midwifery Council (PNMC) — the exact current legal name/statute in force should be confirmed against Pakistan's official gazette given the same period of institutional flux that affected PMDC.			
EMR	Egypt	Egyptian Medical Syndicate (EMS)	Medicine	A semi-governmental professional syndicate founded in	Assists the Ministry of Health in conferring licences	https://en.wikipedia.org/wiki/Egyptian_Medical_Syn	**not verified** (sources could not be opened). Suspect

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				1940, with mandatory membership required for physicians to practise. Sources consulted give conflicting founding/governing law numbers, so the precise statutory citation should be confirmed against the Egyptian Official Gazette before use in the handbo	to physicians; verifies academic credentials on application and issues practice certificates; administers licence renewals and approval of foreign-trained doctors; since February 2021 shares national licensing-exam administration with the newly created Egyptia	dicate; https://grokipedia.com/page/egyptian_medical_syndicate (law citation unverified — flagged)	ed: Law 45/1969 number from Arabic legal database mansh urat.org, not the Official Gazette. Leads: 1 2 3 4

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				ok. (Nursing is regulated separately by its own syndicate — see the row added below.)	n Health Council for new graduates.		
EMR	Egypt	Nursing Professions Syndicate	Nursing	A mandatory professional syndicate for Egyptian nurses, distinct from the Egyptian Medical Syndicate, operating under Law No. 155 of 1976 (Official Gazette,	Membership is mandatory to legally practise nursing in Egypt; the syndicate's Registration Committee verifies nursing credentials and the Ministry of Health-issued practice	Law No. 155/1976; https://nazra.org/en/2013/12/q-about-law-regulation-nursing-professions-syndicate	**not verified** (sources could not be opened). Suspected: Likely typo 155 vs 115; not resolved without the Gazette. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				9 November 1976).	licence before granting membership, with appeal rights through the syndicate's Council and ultimately the Administrative Court.		
EMR	Jordan	Jordan Medical Council (JMC)	Medicine	A body with its own legal personality and financial/administrative independence, headed by the Minister of Health, established under Jordani	Verifies medical credentials of Jordanian and foreign-trained doctors; administers (or grants defined exemptions from) the Jordanian	2022 Jordan Medical Council law, as reported by Jordan Times; https://jordantimes.com/news/local/lower-house-approves-2022-jordani	**not verified** (sources could not be opened). Suspected: Note the JMC itself predates 2022 (earlier Jordan Medical Council Law);

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				an medical council legislation approved by parliament in 2022. Note: the precise official statute number/gazette citation should be verified directly with JMC or the Jordanian Official Gazette before use in the handbook. (Nursing is regulated separately — see the	licensing examination; grants physicians the legal privilege to practise medicine in Jordan; sets category-based registration requirements.	an-medical-council-draft-law; https://datafl owgroup.com/organization/jordan-medical-council/	the 2022 law replaced it - earlier law number not confirmed. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Jordanian Nursing Council row added below.)			
EMR	Jordan	Jordanian Nursing Council (JNC)	Nursing and midwifery	A government institution with administrative and financial independence responsible for regulating the nursing profession in Jordan, first established by temporary legislation in 2002 and formalized permanently	Regulates and advances the nursing and midwifery professions, alongside the separate Jordan Medical Council for physicians; develops nursing human resources and supports scientific research; contrib	Law No. 53 of 2006; https://jnc.gov.jo/En/Pages/JNC_Establishment__	**not verified** (sources could not be opened). Suspected: Core facts consistent. Leads: 1 2 3

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				under Law No. 53 of 2006.	utes to national health-strategy implementation and protects public health.		
EMR	Lebanon	Lebanese Order of Physicians (LOP)	Medicine	The statutory professional order for physicians in Lebanon, operating under Law No. 1658 of 17 January 1979 regulating the practice of the medical profession (Official Gazette,	Determines physicians' professional eligibility to practise medicine; disciplines members against professional and ethical rules; advises government ministries on health-workforce	Law No. 1658 (1979); https://lopbeirut.org/en/law-regulating-the-practice-of-the-medical-profession-in-lebanon/; https://lopbeirut.org/en/lop-role/	**not verified** (sources could not be opened). Suspected: Gazette issue 10 of 8 March 1979 not verified. Leads: 1 2 3

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Issue 10, 8 March 1979). (Nursing is regulated separately — see the Order of Nurses in Lebanon row added below.)	laws and regulations; administers retirement and welfare funds for members.		
EMR	Lebanon	Order of Nurses in Lebanon (ONL)	Nursing	A mandatory regulatory body for Lebanon's nursing profession, established in 2002 and comprising all qualified nurses holding licences	Regulates nursing practice and maintains professional standards; registers and recognizes nurses licensed by the Ministry of Public Health,	Established 2002; https://www.icn.ch/membership/our-members/order-nurses-lebanon-onl	**not verified** (sources could not be opened). Suspected: none Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				issued by the Lebanese Ministry of Public Health; distinct from the Lebanese Order of Physicians, which regulates only medicine.	ensuring only qualified practitioners are authorized to work; works to strengthen patient safety and community health outcomes.		
EMR	Qatar	Department of Healthcare Professions (DHP), Ministry of Public Health	All licensed healthcare practitioners, including medicine and nursing (single omnibus regulator)	The sole regulatory authority for healthcare practitioners in Qatar, created by Emiri Decree No. 14 of 2019 restructuring	Registers and licenses healthcare practitioners as the sole authority for the state; operates a Fitness to Practise functio	Emiri Decree No. 14 (2019); https://dhp.moph.gov.qa/en/Pages/AboutUs.aspx	**not verified** (sources could not be opened). Suspected: Content consistent with official DHP pages. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				the Ministry of Public Health and absorbing the former Qatar Council for Healthcare Practitioners (established by Emiri Decree No. 7 of 2013, building on Law No. 13 of 2002); regulates both medicine and nursing as a single omnibus body.	n for complaints and ethical compliance; sets accreditation standards for medical education and continuing professional development.		
EMR	Saudi Arabia	Saudi Commission for Health	All licensed health professi	A Saudi professional scientific	Classifies and registers health	Royal Order No. M/2 (1992);	**not verified** (sources could

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
		Specialties (SCFHS)	ons, including medicine and nursing (single omnibus regulator)	commission with an independent legal personality, established by Royal Order No. M/2 of 1992 and supervised by a Board of Trustees; functions as a single omnibus regulator covering medicine, nursing, and other health professions rather than separat	practitioners by professional rank through its Professional Practice Executive Board; accredits training programmes and healthcare specialties; sets and enforces continuing professional development and re-licensing requirements.	https://en.wikipedia.org/wiki/Saudi_Commission_for_Health_Specialties	not be opened). Suspected: Sources disagree on exact date (6/2 vs 6/3/1413); SCFHS site favours 6 Safar. Leads: 1 2 3

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				e profession-specific councils.			
EMR	United Arab Emirates (Dubai)	Dubai Health Authority (DHA)	All licensed health professions, including medicine and nursing (single omnibus regulator for Dubai)	A public authority with independent legal personality established under Law No. (6) of 2018 Concerning the Dubai Health Authority (superseding the 2007 law that first created it); one of three UAE healthcare regulat	Authorises individuals and facilities to practise health professions in Dubai and issues professional licences; sets bylaws, standards and requirements for practice; audits and inspects practitioners and facilities for	Law No. (6) of 2018 Concerning the Dubai Health Authority; https://dlp.dubai.gov.ae/Legislation%20Reference/2018/Law%20No.%20(6)%20of%202018%20Concerning%20the%20Dubai%20Health%20Authority.html; https:/	**not verified** (sources could not be opened). Suspected: Gulf News cites it as Law No. 8 - official portal title confirms No. 6. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				ors alongside the federal Ministry of Health and Prevention (MOHAP, Northern Emirates) and the Abu Dhabi Department of Health (DoH) — the UAE has no single national medical or nursing council, and DHA licenses both doctors and nurses as a single omnibu	compliance; investigates complaints and alleged malpractice.	/alkabban.com/medical-law/healthcare-regulatory-authorities-uae/	

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				s regulator for Dubai.			
EUR	Ireland	Medical Council of Ireland	Medicine	Ireland's statutory regulator for the medical profession, originally established by the Medical Practitioners Act 1978 and now operating under the Medical Practitioners Act 2007.	Maintains the publicly searchable register of all medical practitioners in Ireland; sets requirements for undergraduate, intern and postgraduate medical education; makes continuing professional development a legal duty (since 2011); specifie	Medical Practitioners Act 2007; https://en.wikipedia.org/wiki/Medical_Council_of_Ireland	**not verified** (sources could not be opened). Suspected: 2007 Act functions not read from statute text. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandates	Reference	Checked
					s conduct/ethics standards and investigates complaints.		
EUR	Ireland	Nursing and Midwifery Board of Ireland (NMBI)	Nursing and midwifery	The statutory regulatory body for nurses and midwives in Ireland, governed by the Nurses and Midwives Act 2011.	Protects the public and the integrity of the nursing and midwifery professions; maintains the Register of Nurses and Midwives and a Candidate Register for students; sets standards for and approves educati	Nurses and Midwives Act 2011	**not verified** (sources could not be opened). Suspected: Statutory name is Bord Altranais agus Cnáimhseachais na hÉireann (NMBI is the operating name) - not confirmed by fetch. Ministerial-advice

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
					on/training programmes; investigates fitness-to-practise complaints and imposes sanctions; advises the Minister for Health on nursing/midwifery matters.		function not checked. Leads: 1 2
EUR	United Kingdom	General Medical Council (GMC)	Medicine	The statutory regulator for doctors (and, since 2024, physician associates and anaesth	Sets the professional standards doctors must meet; approves and monitors medical educati	Medical Act 1983; https://www.gmc-uk.org/about/what-we-do-and-why/our-mandat	**not verified** (sources could not be opened). Suspected: Note that the AA/PA Order

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				esia associates) in the UK, operating under the Medical Act 1983 as amended, most recently by the Anaesthesia Associates and Physician Associates Order 2024.	on and training; determines eligibility for and maintains the register of licensed doctors (registration plus a licence to practise); investigates fitness-to-practise concerns affecting patient safety.	e; https://en.wikipedia.org/wiki/General_Medical_Council	2024 is a separate instrument for AAs/PAs, not purely an amendment of the Medical Act 1983 (proposed nuance). Leads: 1
EUR	United Kingdom	Health and Care Professions Council (HCPC)	15 allied health and other health/care professi	A statutory corporation and regulatory	Enforces protected professional titles by law;	Health Professions Order 2001	**not verified** (sources could not be opened).

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
			ons (does not cover doctors or nurses/ midwives)	body established under the Health Professions Order 2001 to protect the public across 15 professions (allied health and other health/ care professions; does not cover doctors or nurses/ midwives, who are regulated separately — see the GMC and NMC-	approves qualifying education programs; sets and assesses the baseline Standards of Proficiency and Conduct.		Suspected: Reference lacks the SI number (believed SI 2002/254, not verified by fetch). The current list of 15 professions was not checked item by item. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				UK rows added below).			
EUR	United Kingdom	Nursing and Midwifery Council (NMC)	Nursing, midwifery and nursing associates	The statutory regulator for nurses, midwives and nursing associates across the UK, established under the Nursing and Midwifery Order 2001 (in force April 2002), replacing the former UK Central Council for Nursing, Midwifery and	Maintains the register of nurses, midwives, specialist community public health nurses and nursing associates (over 788,000 registrants as of March 2025); sets and reviews education and training standards; investigates fitness-to-	Nursing and Midwifery Order 2001; https://en.wikipedia.org/wiki/Nursing_and_Midwifery_Council	**not verified** (sources could not be opened). Suspected: Nursing associates are registered for England only (since 2019) - not confirmed by any source found here. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Health Visiting.	practise concerns.		
SEAR	India	Indian Nursing Council (INC)	Nursing	An autonomous statutory body formed by the Government of India under the Indian Nursing Council Act, 1947 (Act No. 48 of 1947), operating under the Ministry of Health & Family Welfare alongside the National Medical	Registers nurses at the national level (including via the Nurses Registration & Tracking System, a live national registry); standardizes nursing curricula; recognizes and inspects nursing education institutions; sets professional-compli	Indian Nursing Council Act, 1947; https://www.indiannursingcouncil.org/uploads/inc-act-1947.pdf; https://testbook.com/ugc-net-paper-1/indian-nursing-council	**not verified** (sources could not be opened). Suspected: No fetch possible. Act No. 48 (XLVIII) of 1947 and the NRTS are consistent with the snippets. Correction needed in 'professions': the INC covers nurses, midwives and health

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Commission.	ance standards.		visitors, not only nursing. Mandate nuance: State Nursing Registration Councils issue the licence to practise, while NRTS enrolment gives a NUID (secondary snippet). Drop testbook.com, which is not an authoritative source. Leads: 1
SEAR	India	National Medical	Medicine (allopathic)	An apex statutory body	Enforces uniform	National Medical	**not verified** (source

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
		Commission (NMC)		constituted by an Act of Parliament known as the National Medical Commission Act, 2019.	standards for undergraduate and postgraduate medical education; maintains the National Register; coordinates with State Medical Councils; administers licensing examinations.	Commission Act, 2019	s could not be opened). Suspected: No fetch possible. The NMC Act 2019 (in force 25 Sept 2020, per a secondary snippet), the National Register and State Medical Councils are consistent with the snippets. 'Administers licensing examinations'

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							overstates current practice, because implementation of the NEXT was not confirmed. No URL is given; add the official Act text or nmc.org.in. Leads: 1
WPR	Malaysia	Malaysian Medical Council (MMC)	Medicine	A statutory council established under Part II of the Medical Act 1971 (Act 50), which consolidated and	Maintains the Register of Medical Practitioners under Part III of the Act; approves qualifications recognized for	Medical Act 1971 (Act 50); https://en.wikipedia.org/wiki/Medical_Act_1971; https://mmc.gov.my/laws-regulations/	**not verified** (sources could not be opened). Suspected: only WebSearch summaries available, so nothing

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				amended the law relating to registration and practice of medical practitioners in Malaysia. (Nursing is regulated separately — see the Nursing Board Malaysia row added below.)	registration; oversees provisional and full registration, including compulsory post-registration public service; enforces professional conduct standards.		quoted from a fetched page. Search leads: Act 50 Part II does establish the MMC; Act still in force but substantially amended in 2012 (definition should mention the amendment). The Wikipedia URL is Wikipedia-only sourcing, so replace it with the MMC-hosted Act

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							PDF. The claim about compulsory public service was not checked. https://mmc.gov.my/laws-regulations/ was not reachable. Leads: 1 2
WPR	Malaysia	Nursing Board Malaysia (Lembaga Jururawat Malaysia)	Nursing (a separate Midwives Board regulates midwifery under the Midwives Act 1966)	Malaysia's statutory nursing regulator, established under the Nurses Act 1950 and the Nurses Registration Regulations	Keeps and maintains the Register of Nurses; regulates nursing practice and training/examination procedures; issues registration	Nurses Act 1950; https://hq.moh.gov.my/nursing/en/about-us/midwives-board/	**not verified** (sources could not be opened). Suspected: Leads support the Nurses Act 1950 (Act 14) and the Nurses Registr

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				1985, operating under the Ministry of Health alongside the separate Malaysian Medical Council.	credentials; enforces professional conduct standards through an established code of ethics.		ation Regulations 1985; give the Act number. The reference URL is the Midwives Board page, not the Nursing Board page, so it is probably the wrong page; the lead suggests hq.moh.gov.my/nursing/?p=3322 (Nursing & Midwives Board). The MOH page reporte

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							dly says the nurses/midwives legislation is under review. Leads: 1 2 [3] (https://www.moh.gov.my/images/04-penerbitan/akta-kesihatan/midviwes act 1966.pdf)
WPR	Singapore	Singapore Medical Council (SMC)	Medicine	A statutory board operating under the Ministry of Health, established under the	Maintains the Register of Medical Practitioners; evaluates and recognizes international degrees	Medical Registration Act 1997 (Chapter 174)	**not verified** (sources could not be opened). Suspected: Leads suggest the 1997 Act

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				Medical Registration Act 1997.	; grants conditional or supervised practice; issues annual practicing certificates.		continued an existing council (the Medical Council of Singapore) rather than newly establishing it, and that SMC is a body corporate; 'statutory board' is unconfirmed. 'Chapter 174' is the pre-2020 Revised Edition citation and may be outdated; this is unconfi

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							rmed, and the search found no source on it. The Act was amended in 2020 (Act 34 of 2020, in force 1 July 2022: disciplinary tribunals moved to a Disciplinary Commission). The reference has no URL; add the SSO or SMC page. Leads: 1 2
WPR	Singapore	Singapore Nursing Board	Nursing and midwifery,	A statutory board	Approves or rejects applica	Nurses and Midwives Act	**not verified** (source

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
		(SNB)	including Advanced Practice Nurses	established under the Nurses and Midwives Act 1999, operating under the Ministry of Health alongside the Singapore Medical Council.	tions for registration/enrolment of nurses and registration of midwives; issues and renews practising certificates; sets and maintains training, education, professional conduct and ethical standards; accredits nursing/midwifery education progra	1999; https://www.snb.gov.sg/who-we-are/functions-of-snb/	s could not be opened). Suspected: The search lead is consistent with the mandate, including APN certification under s.32 of the Nurses and Midwives Act 1999. The SNB site seems to exist on both snb.gov.sg and healthprofessionals.gov.sg/snb, so the reference URL

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
					mmes; certifies Advanced Practice Nurses.		may redirect or be legacy. Its resolution could not be tested. Leads: 1 2
WPR	Australia	Medical Board of Australia (MBA)	Medicine	The statutory National Board responsible for regulating the medical profession in Australia, operating under the Health Practitioner Regulation National Law (2009) as part of Ahpra's	Registers medical practitioners across multiple registration types (general, specialist, provisional, limited, non-practising, student); sets registration standards, codes, guidelines and	Health Practitioner Regulation National Law Act 2009; https://www.medicalboard.gov.au/; https://www.ahpra.gov.au/	**not verified** (sources could not be opened). Suspected: Leads are consistent with the mandate. The statute is the National Law set out in the Schedule to the Health Practitioner Regulat

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
				National Registration and Accreditation Scheme.	CPD requirements; manages notifications/complaints; assesses overseas-trained doctors.		ion National Law Act 2009 (Qld) (Act 45 of 2009), adopted by each state and territory; the citation should make clear it is the Queensland host Act. The registration-types list could not be checked. Neither reference URL was reachable. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
WPR	Australia	Nursing and Midwifery Board of Australia (NMBA)	Nursing and midwifery	A statutory National Board established under the Health Practitioner Regulation National Law, working in partnership with Ahpra.	Registers practitioners and students; develops uniform national professional standards, codes, and guidelines; approves accredited programs of study.	Health Practitioner Regulation National Law Act 2009	**not verified** (sources could not be opened). Suspected: Leads are consistent with the mandate: NMBA approves programs and ANMAC accredits them. The reference has no URL; add nursingmidwiferyboard.gov.au. Leads: 1 2
WPR	New	Medica	Medici	A	Registe	Health	**not**

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
	Zealand	l Council of New Zealand (MCNZ)	ne	statutory authority appointed and operating under the Health Practitioners Competence Assurance Act 2003.	rs medical practitioners; sets scopes of practice and competence standards; issues practicing certificates; reviews performance and fitness to practice.	Practitioners Competence Assurance Act 2003	**verified** (sources could not be opened). Suspected: Leads are consistent: MCNZ is the 'responsible authority' (the HPCAA term; 'statutory authority appointed' is loose wording). The reference has no URL; add mcnz.org.nz or legislation.govt.nz. Leads: 1 2

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
WPR	New Zealand	Nursing Council of New Zealand (NCNZ)	Nursing	New Zealand's statutory regulator for nursing, first established in 1902 (making New Zealand the first country to legally mandate nurse registration) and now operating under the Health Practitioners Competence Assurance Act 2003.	Registers nurses and sets standards for nursing education, competence and practice; issues annual practising certificates; investigates fitness-to-practise concerns, in parallel with the Medical Council of New Zealand's equivalent role for doctors.	Health Practitioners Competence Assurance Act 2003; https://en.wikipedia.org/wiki/Nursing_Council_of_New_Zealand	**not verified** (sources could not be opened). Suspected: Leads say the Nurses Registration Act 1901 took effect on 1 January 1902 and was administered by the Health Department, so it is doubtful that the Council itself was 'established in 1902'. The 1902

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							date and the claim that New Zealand was the first country appear to come from Wikipedia and are unconfirmed. NCNZ also regulates enrolled nurses and nurse practitioners (unchecked). The reference is Wikipedia-only for the history; replace it with nursing

WHO region	Country	Council	Regulates	Statutory definition	Mandate	Reference	Checked
							council.org.nz or NZHistory. Leads: 1

Professional councils and boards: statutory definitions and mandates (AI generated in the workbook, then fact-checked; see the Checked column)

Data element terms

Table.

Term	Description
Identification number	[Definition to draft]
Name	[Definition to draft]
Birth history	[Definition to draft]
Sex	Male or Female, based on physical characteristics observed at birth. [Provisional definition, NHWA handbook 2nd ed.]
Citizenship and language	[Definition to draft]
Address	[Definition to draft]
Contact information	[Definition to draft]
Occupation	The concept of occupation is defined as a “set of jobs whose main tasks and duties are characterized by a high degree of similarity”. A job is defined as “a set of tasks and duties performed, or meant to be performed, by one person, including for an employer or in self-employment”. The National Health Workforce Accounts covers health and health-related occupations grouped according to the International Standard Classification of Occupations 2008. [Provisional definition, NHWA handbook 2nd ed.]
Professional licence and certification	[Definition to draft]

Term	Description
Employment status	[Definition to draft]
Employment address	[Definition to draft]
Data submission institution	[Definition to draft]
Activity level	• Practising: This includes all health and care personnel directly providing services to or for patients and communities. • Professionally active: This includes practising health and care personnel as well as other personnel who are directly not providing services to patients but for whom their medical or paramedical education is a prerequisite for the execution of the job (e.g. education, research, public administration). • Licensed to practice: This includes all health and care personnel who are registered and entitled to practice. [Provisional definition, NHWA handbook 2nd ed.]
Subnational unit of practice	To be defined according to the specific conditions, governing structures, and constitutional provisions existing in a given country. Disaggregation based on administrative boundaries down to the first or second subnational level is recommended (depending on the structure of administrative boundaries and the size of subnational territories), without overlaps between the administrative units. Examples for subnational administrative units are states, regions, provinces, counties, and districts. [Provisional definition, NHWA handbook 2nd ed.]
Facility ownership	Classification for ownership type: • Publicly-owned: Facilities owned or controlled by a governmental unit or a public corporation (where control is defined as the ability to determine the general corporate policy) corresponding to the institutional sector “government units” defined in

Term	Description
	the System of National Accounts 2008. • Not-for-profit privately-owned: Facilities that are legal or social entities created for the purpose of producing goods and services, whose status does not permit them to be a source of income, profit, or other financial gain for the unit(s) that establish, control or finance them, corresponding to the institutional sector “non-profit institutions serving households” defined in the System of National Accounts 2008. • For-profit privately-owned: Facilities that are legal entities set up for the purpose of producing goods and services and are capable of generating a profit or other financial gain for their owners corresponding to the institutional sector “non-financial corporations” defined in the System of National Accounts 2008. [Provisional definition, NHWA handbook 2nd ed.]
Facility type	Facility types are categorized following the system of health accounts through the classification of health care industries. These enable arrangement of country-specific institutions in common, internationally applicable categories. The categories are as follows: Hospitals (HP.1) Hospitals comprise licensed establishments that are primarily engaged in providing medical, diagnostic and treatment services that include physician, nursing and other health services to inpatients and the specialised accommodation services required by inpatients. Hospitals provide inpatient health services, many of which can be delivered only by using specialised facilities and professional knowledge as well as advanced medical technology and equipment,

Term	Description
	which form a significant and integral part of the provision process. Although the principal activity is the provision of inpatient medical care they may also provide day care, outpatient and home health care services. The tasks of hospitals may vary by country and are usually defined by legal requirements. In some countries, health care facilities need in addition a minimum size (such as a number of beds and medical staff to guarantee 24-hour access) in order to be registered as a hospital. Residential long-term care facilities (HP.2) The category of Residential long-term care facilities comprises establishments that are primarily engaged in providing residential long-term care that combines nursing, supervisory or other types of care as required by the residents. In these establishments, a significant part of the production process and the care provided is a mix of health and social services, with the health services being largely at the level of nursing care, in combination with personal care services. The medical components of care are, however, much less intensive than those provided in hospitals. Providers of ambulatory health care (HP.3) This item comprises establishments that are primarily engaged in providing health care services directly to outpatients who do not require inpatient services. This includes both offices of general medical practitioners and medical specialists and establishments specialising in the treatment of day-cases and in the delivery of home care services. Health practitioners in ambulatory health care primarily provide services to

Term	Description
	patients who visit the health professional's office, or the practitioners visit the patients at home. Consequently, these establishments do not usually provide inpatient services. Ancillary services (HP.4, including transportation, emergency rescue, laboratories and others) This category comprises establishments that provide specific ancillary type of services directly to outpatients under the supervision of health professionals and not covered within the episode of treatment by hospitals, nursing care facilities, ambulatory care providers or other providers. Included are providers of patient transportation and emergency rescue, medical and diagnostic laboratories, dental laboratories and other providers of ancillary services. These specialised providers may charge patients directly for their services rendered or may provide these ancillary services as benefits-in-kind under special service contracts. Retailers (HP.5, including pharmacies) This item comprises specialised establishments whose primary activity is the retail sale of medical goods to the general public for individual or household consumption or utilisation. Establishments whose primary activity is the manufacture of medical goods, such as making lenses, orthopaedic or prosthetic appliances for direct sale to the general public for individual or household use, are also included, as is fitting and repair done in combination with sale. Note: Due to special medical safety and quality regulations, retailers of over-the-counter medical products and other providers of medical goods are subject

Term	Description
	to licensing and/or pharmaceutical authorisation in order to be eligible to provide their activities. Non-health care products such as cosmetics, dietetic products and natural products are excluded from health expenditures. Providers of preventive care (HP.6) This category comprises organisations that primarily provide collective preventive programmes and campaigns/public health programmes for specific groups of individuals or the population-at-large, such as health promotion and protection agencies or public health institutes as well as specialised establishments providing primary preventive care as their principal activity. This includes the promotion of healthy living conditions and lifestyles in schools by special outside health care professionals, agencies or organisations. [Provisional definition, NHWA handbook 2nd ed.]
Place of training	This is defined as the place of first qualification for the health worker. Personnel may be categorized as either domestic-trained, foreign-trained, or their place of training may be unknown. The foreign-trained health personnel may also be disaggregated into “foreign-trained national-born” (people born in a country who went to study in another country but have come back afterwards to practice in their home country) and “foreign-trained foreign-born”. [Provisional definition, NHWA handbook 2nd ed.]
Date of entry into the active workforce	[Definition to draft]
Date and type of exit from the active workforce	[Definition to draft]
Type of contract	The contract type is a key

Term	Description
	employment characteristic of all workers, including health workers. Personnel may be categorized as full-time workers or part-time workers, depending on their specific contract types. Part-time workers refers to those employed persons whose normal hours of work (calculated weekly or on average over a given period of employment) are less than those of comparable full-time workers. [Provisional definition, NHWA handbook 2nd ed.]
Qualification and education programme	[Definition to draft]
Training institution and year of graduation	[Definition to draft]
Employer sector	Relevant sectors based on the System of National Accounts 2008: government units; non-profit institutions serving households; non-financial corporations; and households. [Provisional definition, NHWA handbook 2nd ed.]
Salary grade at entry	[Definition to draft]
Leadership position	A senior government official position (e.g., Commissioner, Director-General, Director, etc.) at the national level in the Ministry of Health [Provisional definition, NHWA handbook 2nd ed.]

Data element terms: one per data element in chapter 11. Provisional definitions are quoted from the NHWA handbook, 2nd edition, for working-group review

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